

Takings for Economic Development: An Empirical Analysis of Eminent Domain Exercises in New York City*

Ronit Levine[†] Gary A. Wagner[‡]

September 2026

Abstract

Despite extensive debate over eminent domain for economic development, empirical evidence on the actual economic effects of condemnations at the parcel level remains scarce. Existing studies rely on state-level aggregates or regulatory proxies that cannot capture localized impacts where condemnations actually occur, leaving the parcel-level economic effects of takings across an entire city unexamined. We address this gap by analyzing a novel hand-coded dataset of all takings in New York City from 1991 to 2019, merged with geocoded establishment-level data on employment and business counts, parcel-level assessed land and building values, and building permit issuances. A spatial externality framework derived from the monocentric city model generates testable predictions that discipline our outcome selection and spatial research design. We employ partially pooled synthetic control methods to construct credible counterfactuals, with treatment and comparison groups defined by proximity to project parcels to assess sensitivity to spatial boundary choices. Across these outcomes and specifications, we find limited and inconsistent evidence of positive economic spillovers from condemnation projects. Where statistically significant effects emerge, they are more frequently negative than positive, with significant positive estimates confined to building permit issuances and significant negative estimates spread across the remaining outcomes. These findings raise questions about the empirical foundations of the spillover benefit rationale commonly invoked to justify development-oriented condemnations, while acknowledging that our analysis is representative of one jurisdiction and should be interpreted accordingly.

Keywords: economic development, urban renewal, *Kelo*, eminent domain

JEL Classification Numbers: R51, H76, R11

*The authors would like to thank the editors, two anonymous referees, and Jonathan Murphy for several comments that have improved the paper. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Any errors are our own.

[†]Faculty of Law, Tel Aviv University. Email: rls@tauex.tau.ac.il

[‡]Acadiana Business Economist Endowed Chair and Professor of Economics, University of Louisiana at Lafayette. Email: gary.wagner@louisiana.edu.

1 Introduction

In *Kelo v. City of New London* (2005) the U.S. Supreme Court reaffirmed that the Public Use Clause of the U.S. Constitution Fifth Amendment does not deny states the power to take private property for “public purpose” that is not a collective good. The Court held that taking may be constitutional even if the property is not blighted and even when the land will be designated for private uses such as housing or commercial purposes.¹ The *Kelo* decision therefore permitted the forced transfer of private property to private parties for the purpose of generating anticipated economic development.

The city of New London, whose taking decision was at the focus of the *Kelo* case, suffered from severe unemployment rates (nearly double that of the state), and from ongoing loss of residents which brought its population to its lowest level in 80 years. Pfizer Inc.’s announcement that it would build a \$300 million research facility in the town led New London’s local officials to hope that it would “draw new business to the area, thereby serving as a catalyst to the area’s rejuvenation” (*Kelo v. City of New London* 2005, at 473). The city thereafter formulated an integrated development plan that included the taking of private properties, in the belief that the plan would provide appreciable benefits to the community and capitalize on the arrival of the Pfizer facility. The main economic objectives were creating jobs and generating tax revenue.

In rejecting the property owners’ petition against their land taking, the Court ruled that “[p]romoting economic development is a traditional and long-accepted function of government” (*ibid.*, at 484). Relying on the “comprehensive character” of the development plan adopted by the city, the Court thought it appropriate to resolve the challenges posed by the individual owners in light of the entire plan. The majority contended that “[b]ecause that plan unquestionably serves a public purpose, the takings challenged here satisfy the public use requirement of the Fifth Amendment.”

In her dissenting opinion, Justice O’Connor noted that the Court’s decision significantly expands the meaning of public use, holding that private property “currently put to ordinary private use” may be taken so long as the new use is *predicted* to generate some secondary benefits for the public, such as increased tax revenue, more jobs, or even aesthetic pleasure (*ibid.*, at 501). Justice

¹Taking for urban renewal that includes the destruction of deteriorated housing is also known as “blight condemnation.” However, the exact definition for blight is subject to interpretation and is not standardized.

O'Connor thus makes it clear that nothing in the Court's decision prohibits property transfers "whose only projected advantage is the incidence of higher taxes, or that hope to transform an already prosperous city into an even more prosperous one" (*Kelo v. City of New London* 2005, at 504).

Consequently, the Supreme Court delegated to local governments the authority to seize private property for economic development based entirely on their own forecasts of anticipated economic benefits. While courts have inherent limitations in their capacity to meaningfully evaluate planning decisions or economic forecasts, this raises the question: are local governments actually equipped to make reliable economic predictions?

Our goal is not to identify a flaw in *Kelo*'s reasoning. There are indeed good reasons for courts to allow deference to local governments in setting public purposes and to refrain from "second guess[ing] the City's considered judgments about the efficacy of its development plan" (*ibid.*, at 488). We contend that regardless of whether local governments should possess broad authority in eminent domain decisions, and regardless of whether promoting local economic development can justify transferring property from one private party to another, there remains a fundamental need for empirical evidence demonstrating that such economic projections are well-grounded. The existing literature provides a helpful foundation, but there are opportunities to strengthen the evidence along multiple dimensions. Studies using actual takings data operate only at the state level, making it impossible to detect localized economic effects. Research using regulatory measures as proxies offers useful insights but does not directly assess the effects of actual condemnations.

This study addresses these gaps by providing a more comprehensive empirical examination of eminent domain's actual economic effects using parcel-level data. Whereas Levine-Schnur and Wagner (2025) offer an abbreviated treatment of the economic evidence as part of a broader normative argument, the present paper advances a formal spatial externality framework derived from the monocentric city model, derives testable predictions for employment, establishment counts, and property value capitalization, and tests those predictions using a rigorous quasi-experimental identification strategy and a spatial research design capable of detecting effects across multiple geographic scales. We make four key contributions to the literature. First, we analyze actual condemnation exercises at the local level using an original hand-coded dataset of all takings in New York City from 1991 to 2019 (Levine-Schnur 2023). Second, we employ establishment-level eco-

economic data to track employment, business establishment counts, assessed land and building values, and building permit issuances at the precise locations where condemnations occurred. Third, we implement a rigorous identification strategy using partially pooled synthetic control methods that construct credible counterfactuals for what would have happened without condemnation. Fourth, we systematically examine economic outcomes across multiple treatment and comparison group definitions, with comparison groups drawn from areas up to 800 meters from project sites, to assess robustness to spatial boundary choices.

To conduct this analysis, we merge our takings dataset with business-level data from the geocoded Your Economy Time Series database, allowing us to track changes in employment and business establishment counts in areas surrounding condemnation sites. We supplement this with parcel-level assessed land and building values from MapPLUTO and geocoded building permit issuances from the New York City Department of Buildings to capture planned construction activity and the capitalization of any economic amenities into property values.

To isolate the causal effect of takings, we compare parcel-level economic outcomes in and around taking locations to nearby parcels in New York City where no takings for economic development (or any other purposes) occurred. We estimate counterfactual parcel outcomes (what would have happened in the absence of the taking) using the recently developed partially pooled synthetic control (PPSC) method (Ben-Michael et al. 2022). Like traditional synthetic control methods, the PPSC approach constructs counterfactual outcomes using a weighted average of New York City parcels that have never experienced takings (since 1990), but it is specifically adapted for panel data settings with staggered treatment timing.

The spatial research design is motivated by a formal theoretical framework developed in Section 2.2, which adapts the monocentric city model to generate predictions about how a taking-induced amenity should affect surrounding parcels. The framework predicts that any positive spillover from a condemnation site should attenuate with distance, capitalize into assessed land values at nearby parcels, and manifest in both the intensive margin (employment per firm) and the extensive margin (firm entry and exit). These predictions discipline both the choice of outcomes and the spatial structure of the research design. To ensure robust results and capture potential spillover effects consistent with these predictions, we define “treated” areas using four different lot-level specifications that progressively expand the geographic scope where effects might be observed.

For each alternative specification, we estimate synthetic counterfactuals for employment, business establishment counts, assessed land and building values, and building permit issuances using four distinct comparison groups of unaffected properties located near the condemnation sites, yielding 16 specifications per outcome.

Across all 160 specifications, the evidence does not provide consistent support for the view that eminent domain exercises generate positive economic spillovers. A negative point estimate indicates that condemnation is estimated to lower the outcome relative to its synthetic control counterfactual, the opposite of the increases in employment, business activity, and property values predicted by the economic development rationale. Such estimates predominate: 111 of 160 (69.3%) are negative across all outcomes and project types. The positive estimates are concentrated almost entirely in building permit issuances, an outcome that captures the city’s approval of proposed construction plans rather than realized economic activity. Statistically significant negative estimates outnumber statistically significant positive ones by a margin of 33 to 17, with the significant positive estimates limited to permit issuances and the significant negative estimates spread across employment, establishment counts, and assessed property values. Assessed building values are particularly consistent: for non-urban renewal condemnations, all 16 specifications yield statistically significant negative estimates, and for urban renewal projects estimates for both land and building values are uniformly significant and negative across multiple specifications. Taken as a whole, the evidence offers limited empirical support for the spillover benefit rationale underlying development-oriented condemnations, though we caution that our analysis is representative of New York City and the findings should be interpreted with appropriate modesty regarding generalizability to other contexts.

We continue as follows. Section 2 reviews the literature on takings for economic development and economic outcomes, and presents a theoretical framework to guide the empirical analysis. Section 3 describes our research question, datasets, and methods. Results are presented in Section 4, while Section 5 discusses limitations and directions for future research.

2 Background and Theoretical Framework

2.1 Literature Review

In *Kelo v. City of New London* the Court held that “It would be incongruous to hold that the City’s interest in the economic benefits to be derived from the development...has less of a public character...than any of those other interests,” and that “there is no basis for exempting economic development from our traditionally broad understanding of public purpose.” (*Kelo v. City of New London* 2005, at 485). However, this justification represents what Miceli and Segerson (2007) term the “ends approach” to public use, focusing on the ultimate purpose of the taken land rather than the economic mechanism that necessitates government intervention. The economic literature suggests that the proper justification for eminent domain lies not in spillover benefits per se, but in addressing the “holdout problem” during land assembly, where individual property owners can exercise monopoly power to extract prices far above their true valuations, potentially blocking efficient development projects entirely (Merrill 1986).

This theoretical distinction is important because it implies that eminent domain should be available for any project requiring assembly of multiple parcels, whether public or private, rather than being limited to projects with demonstrable public benefits. The holdout problem creates a form of Prisoner’s Dilemma where landowners prefer to delay sales hoping to extract higher prices, leading to socially inefficient outcomes even when projects would generate net benefits (Miceli and Segerson 2007). Miceli and Segerson’s (2021) analysis indicates that there exist conditions under which using eminent domain for private development would increase efficiency, but equally importantly, there are conditions under which its use would be welfare-reducing.² They show that a necessary condition for welfare improvement is a pre-existing market imperfection in housing and/or labor markets. Without such distortions, markets lead to efficient transactions, and government interference necessarily reduces social welfare.

The theoretical literature also reveals a fundamental tension in compensation policy. The seminal work by Blume et al. (1984) demonstrates that paying full compensation creates moral hazard problems; landowners will overinvest in property subject to taking risk, knowing they will

²See also Glaeser’s opinion that “The use of tax breaks and eminent domain to lure large employers to declining cities is almost always an error.” Edward Glaeser, They Can Take It If They Want It, Wall Street Journal, Jul. 24, 2015, <http://www.wsj.com/articles/they-can-take-it-if-they-want-it-1437767638> (last visited Jan 25, 2025).

be compensated for losses. This suggests zero compensation would be efficient, but constitutional requirements and fairness concerns necessitate some payment. Various solutions have been proposed, including “threshold rules” that provide compensation only for inefficient takings (Miceli and Segerson 1994) and recognition that compensation financed through property taxes may have offsetting budgetary effects through the government budget constraint (Fischel and Shapiro 1989).

Empirically testing these theoretical predictions is, however, a very complicated task, first and foremost because of the lack of available data on actual takings: “eminent domain has proven difficult to study due to severe data constraints” (Kitchens 2014, p. 456). The limited empirical literature has employed two main approaches: studies using actual takings data and those using regulatory standards as proxies for eminent domain activity. Kerekes and Stansel (2016) provide, to the best of our knowledge, the only attempt to test the spillover benefits theory based on actual numbers of eminent domain takings. However, their analysis is conducted at the state level only, using data from the Institute for Justice comparative review of states’ takings practices (Berliner 2003, 2006). They find no evidence that the number of takings for economic purposes at the state level increases tax revenue in a state but do find some evidence of a negative relationship between takings and future tax revenue growth. The empirical findings from the Institute for Justice studies have been relied upon by several other papers as well, although not for the specific question of economic spillover effects (Kerekes 2011; Lanza et al. 2013; Lopez et al. 2009; Sharp and Haider-Markel 2008).

Other studies have employed comparative regulatory standards to test economic metrics rather than actual takings data. For instance, Turnbull and Salvino (2009) measure eminent domain using a dummy variable that takes the value of one for states that explicitly empower local governments to use eminent domain for private development projects. They found that eminent domain for private benefit is associated with higher levels of tax revenue and own-source revenue as a percentage of personal income. However, subsequent research using similar regulatory approaches has yielded mixed or negative results. Carpenter and Ross (2010) found no evidence that restrictions on takings impact state-level construction jobs, building permits, and property taxes. Byrne (2017) examined the impact of restrictions on gross state product and total state employment using panel data, finding no adverse effects on these measures. Turnbull et al. (2017) found a negative and significant relationship between the regulatory availability of eminent domain for private economic

development and growth in private sector employment. Most recently, Bryne (2024) found that effective restrictions on economic development projects at the regulatory level do not have a statistically significant impact on metropolitan statistical area employment or earnings. These results support the view of opponents to takings for economic development purposes: allowing local governments to solve the holdout problem through eminent domain takings does not improve economic efficiency.

Relatedly, Chen and Yeh (2023) examined the effects of pro-taking court precedents rather than direct takings activity. They reported that court decisions that expanded the government’s power of expropriation spurred economic growth and property prices by 0.2 percent but reduced racial minorities’ home ownership and employment by 0.5 percent and 0.3 percent, respectively. Their study did not distinguish, however, between pro-taking decisions regarding takings for strictly public purposes (roads, public spaces, public buildings, etc.) and decisions regarding economic development purposes.

Recent scholarship has highlighted the dynamic nature of eminent domain regulation over time. Lopez et al. (2025) argue that the effectiveness of post-*Kelo* reforms may diminish through a process of “loophole mining,” where developers and local authorities find increasingly creative ways to circumvent restrictions as public scrutiny wanes. Drawing on the economic principle that demand curves become more elastic over time as participants develop substitutes, they suggest that initial regulatory restrictions naturally have a “half-life” as technological and economic barriers are overcome. Their comparative case study of Michigan and North Carolina reveals this dynamic in practice, suggesting that the effectiveness of regulatory approaches cannot be evaluated solely on their initial restrictiveness, but must account for long-term adaptive responses by interested parties.

The economic literature has also evolved to include experimental research providing important insights into the underlying coordination problems that eminent domain is intended to address. Isaac et al. (2016) use laboratory experiments to examine whether site-development competition can facilitate land assembly without government intervention, finding that competition among prospective sellers more than doubles aggregation rates compared to baseline scenarios. Their results suggest that market-based mechanisms may be more effective than previously thought at overcoming holdout problems, particularly when developers can choose between dealing with consolidated versus fragmented land holdings. Portillo (2019) extends this experimental approach to examine

how positive post-development externalities affect land assembly outcomes. His findings reveal that externalities can actually reduce aggregation rates when buyers cannot fully capitalize the externality into their offers, creating a new source of market failure distinct from the traditional holdout problem. However, when transaction surplus is sufficient to cover externality effects, landowners can capture approximately 65% of the externality value and aggregation rates improve substantially. This research highlights the complexity of coordination problems in land assembly and suggests that the effectiveness of both private negotiation and government intervention may depend critically on the specific nature of development externalities.

Despite this growing body of research, little existing work has examined the economic effects of condemnations for economic development projects at the city level using direct data on takings rather than regulatory standards. The closest antecedent is Levine-Schnur and Wagner (2025), which provides descriptive difference-in-differences estimates of employment and establishment effects as part of a broader normative argument; the present paper extends that work by examining a more comprehensive set of outcomes, employing a formal spatial externality framework to discipline outcome selection and research design, and implementing a rigorous quasi-experimental identification strategy that constructs credible counterfactuals and accounts for staggered project timing.

We argue that such analysis is urgently needed, as local-level takings and economic data are indispensable to understanding the true local economic effects of taking activity. An important question in this context is the theoretical geographical scope of any anticipated economic effects, which our city-level analysis is uniquely positioned to address.

2.2 Theoretical Framework

We propose a theoretical framework to motivate the empirical design outlined in the next section. The framework adapts the monocentric city model tradition (Alonso 1964; Mills 1967; Muth 1969) to micro-geographic amenities generated by land use changes at taking sites. In its classical form, the monocentric city model describes an urban area organized around a single center where employment concentrates and to which residents commute. Locations nearer the center economize on commuting and transport costs, so they command higher land rents, and land values fall in a predictable gradient with distance from the center. The lesson we carry into the micro-

geographic setting is that a fixed local source of economic value capitalizes into surrounding land prices and that this benefit attenuates with distance.

The legal justification for economic development takings under *Kelo* rests implicitly on a complementarity assumption: that condemned land, once redeveloped, generates spillover benefits for surrounding parcels sufficient to justify the costs of displacement. As the Court notes in *Kelo*, city governors’ determined that the Fort Trumbull area “was sufficiently distressed to justify a program of economic rejuvenation.” (*Kelo v. City of New London* 2005, at 454). The framework developed here makes that assumption explicit. The key prediction is that any positive economic complementarity from a taking should attenuate with distance from the site, providing a direct mapping between the *Kelo* rationale and the empirical analysis that follows.

Consider a taking at location s that transforms the land use from its prior use to a new use u (infrastructure, public facility, commercial redevelopment, etc.). This transformation may generate a local amenity $\phi(s, u) \geq 0$ that affects the productivity and value of nearby parcels.

A representative firm at parcel i , located at distance d_i from the taking site s , produces output according to:

$$Y_i = A_i \cdot f(K_i, L_i) \cdot \Gamma(d_i) \quad (1)$$

where A_i is parcel-specific total factor productivity, $f(\cdot)$ is a standard production function with capital K_i and labor L_i , and $\Gamma(d_i)$ captures the degree to which the taking raises local productivity through agglomeration spillovers:

$$\Gamma(d_i) = 1 + \phi(s, u) \cdot g(d_i) \quad (2)$$

The function $g : \mathbb{R}_+ \rightarrow [0, 1]$ governs spatial decay and is assumed to satisfy three properties: $g(0) = 1$, so that the maximum benefit is at the location of the taking; $g'(d) < 0$, so that the benefit monotonically declines with distance; and $\lim_{d \rightarrow \infty} g(d) = 0$, so that the amenity dies out with distance. If a taking does not generate an amenity ($\phi = 0$), or if a parcel is far enough from the taking site that the amenity has fully attenuated ($d_i \rightarrow \infty$), then $\Gamma(d_i) \rightarrow 1$ and the production function reduces to its baseline form: the taking generates no productivity effect on that parcel.

In a spatial equilibrium, the amenity capitalizes into land values. Suppose the owner of parcel i

earns a flow of profits $\pi(\Gamma(d_i))$ from operating on the parcel, where $\pi(\cdot)$ is the profit function. The value of the parcel is the present discounted value of this flow:

$$V_i = \frac{\pi(\Gamma(d_i))}{r} \quad (3)$$

where $r > 0$ is the discount rate (assumed constant for simplicity).

Before the taking, the amenity is absent and $\Gamma = 1$. After the taking, $\Gamma(d_i) = 1 + \phi(s, u) \cdot g(d_i) \geq$

1. The change in land value for parcel i is:

$$\Delta V_i = \frac{\pi(\Gamma(d_i)) - \pi(1)}{r} \quad (4)$$

Since π is increasing in Γ and Γ is decreasing in d_i , it can be shown that:

$$\frac{\partial \Delta V_i}{\partial d_i} < 0 \quad (5)$$

That is, the property value impact of the taking declines monotonically with distance from the site. Parcels immediately adjacent to the taking benefit most and the benefit attenuates continuously as distance increases and eventually becomes zero.

In addition to land values, the amenity may also affect employment along the extensive and intensive margins. Consider first the intensive margin. A profit-maximizing firm hires labor up to the point where the value of the marginal product equals the wage, $p \cdot \text{MPL}(L_i, \Gamma) = w$. When Γ rises, the marginal product of labor increases, inducing the firm to expand its labor demand. Standard factor demand therefore implies:

$$\frac{\partial L_i^*}{\partial \Gamma} > 0 \quad (6)$$

Since $\Gamma(d_i) = 1 + \phi(s, u) \cdot g(d_i)$ and $g'(d) < 0$ by assumption, the amenity itself attenuates with distance from the taking site. It follows that optimal employment also declines with distance:

$$\frac{\partial L_i^*}{\partial d_i} = \frac{\partial L_i^*}{\partial \Gamma} \cdot \underbrace{\frac{\partial \Gamma}{\partial d_i}}_{< 0} < 0 \quad (7)$$

so that the employment benefit, like the amenity itself, attenuates monotonically with distance from the taking location.

This framework also provides a mechanism through which takings can affect employment via the extensive margin (whether new firms enter). Suppose a potential entrant at parcel i pays a fixed cost F to operate and earns expected profits:

$$\pi_i = \Gamma(d_i) \cdot \bar{\pi}(A_i) - F \quad (8)$$

where $\bar{\pi}(A_i)$ is the baseline expected profit given location fundamentals. A firm enters if $\pi_i \geq 0$, which requires:

$$\Gamma(d_i) \geq \frac{F}{\bar{\pi}(A_i)} \quad (9)$$

Since Γ declines with distance, there exists a critical threshold d^* beyond which the amenity is too weak to induce entry of marginal firms or prevent exit of incumbents. The number of active establishments at distance d can be written as:

$$N(d) = N_0(d) + \Delta N(d) \quad (10)$$

where N_0 is the baseline count and $\Delta N(d) \geq 0$ is net entry induced by the amenity, with $\Delta N'(d) < 0$.

Three cases arise depending on the relationship between fixed costs and baseline profitability. If fixed costs are low relative to baseline profits ($F < \bar{\pi}(A_i)$), then the firm would enter even in the absence of the amenity. The critical distance is effectively infinite, and the taking induces no additional entry because the marginal entrant is already profitable without it. At the other extreme, if fixed costs are so high that $F > \bar{\pi}(A_i)(1 + \phi)$, the amenity is too weak to cover entry costs even at the taking site itself, and no entry occurs at any distance. The empirically relevant case lies between these two extremes: when $\bar{\pi}(A_i) \leq F \leq \bar{\pi}(A_i)(1 + \phi)$, the amenity is strong enough to induce entry close to the taking site but not far from it, and a well-defined critical distance d^* separates the region of induced entry from the region where the amenity is too attenuated to matter.

To summarize, the model yields three sets of testable predictions. First, if the taking generates a positive amenity ($\phi > 0$), assessed property values at nearby parcels should rise, with the effect

attenuating with distance. Second, on the intensive margin, existing firms should expand employment in response to the productivity gain, again with attenuation. Third, on the extensive margin, the entry condition implies that new firms should be drawn in up to the critical distance d^* , beyond which the amenity is too weak to cover fixed entry costs. Under the null of no complementarity from a taking ($\phi = 0$), all three predictions collapse to zero at every distance.

These predictions are parallel but distinct, and the margins can move independently. A taking that generates foot traffic may attract new retail entrants without substantially affecting the labor demand of existing firms. Conversely, a taking that creates a transit node may raise productivity of incumbents and induce employment expansion without drawing in new entrants (Arzaghi and Henderson 2008). Finding effects on one margin but not the other would therefore be informative about the channel through which the amenity operates. Null results across all three outcomes and all distances would indicate that the taking generates no detectable amenity through any of the channels the model identifies.

3 Data & Methods

3.1 New York City’s Taking Dataset

Following *Kelo* all but five states enacted reforms to their eminent domain laws or constitutions (Berliner 2015).³ Most of these legislative reforms took place soon after the decision (Somin 2015). New York is one of the few states that has not reformed their eminent domain law. Moreover, the State’s highest court reapproved the pro-taking practices even after *Kelo* (*Goldstein v. New York State Urban Development Corp.* 2009; *Kaur v. New York State Urban Development Corp.* 2010; Mihaly and Smith 2011), making an exception to the general post-*Kelo* pattern of stronger state judicial scrutiny of public use issues (Somin 2015, pp. 190–191). Noticeably, in Oklahoma, one of the five states that did not reform their takings laws following *Kelo*, the high courts increased protections against takings for private use (*ibid.*, pp. 182–183). Based on the case law and scholarship (Somin 2010), New York seems to be a case of resistance to the “*Kelo* storm.”⁴ This regulatory stability makes New York an ideal case study for analyzing the economic effects of

³Berliner (2003) counts only 44 states making statutory or constitutional improvements, although she acknowledges that the 45th state, New Jersey, did pass a statute purporting to increase eminent domain protections. For an up-to-date listing of post-*Kelo* state legislative initiatives, see, The Castle Coalition: 50 State Report Card: Tracking Eminent Domain Reform Legislation since *Kelo*, <http://castlecoalition.org/50-state-report-card>.

⁴The other states are Arkansas, Hawaii, and Massachusetts. Additionally, no amendments took place in the District of Columbia, and the U.S. territories.

takings for economic development projects over an extended period following the *Kelo* decision (see Levine-Schnur and Wagner 2025). It should be emphasized that eminent domain processes vary substantially across jurisdictions in terms of political oversight, planning capacity, and project selection. New York City’s institutional context is documented in detail elsewhere (see Levine-Schnur 2023), and readers should bear in mind that our findings are representative only of this jurisdiction when assessing the generalizability of the results.

However, as documented by Levine-Schnur (*ibid.*), the *Kelo* decision did affect political behavior, but in the opposite direction than commonly believed. Analysis of taking exercises in New York City between 1991 and 2019 revealed that the probability of a taking for economic development or urban renewal dropped to just 10% of pre-2005 levels. The use of eminent domain for such projects declined dramatically even though both state and federal courts refrained from imposing any legal limits (*ibid.*).

In New York City, each property’s unique location is identified by its BBL: Borough, Block, Lot. The “lot” refers to a Tax Lot, which is a subdivision of a Tax Block, and a Tax Block is a subdivision of the borough containing real properties (New York City 2007). Between 1991 and 2019, the city exercised its condemnation powers over 197 projects encompassing 3,359 lots and 640 blocks across the five boroughs. Our dataset of 3,359 condemned lots represents approximately 0.4% of the city’s nearly 840,000 total lots (Ellen et al. 2016).

For each of the 197 projects, the project purpose was determined based on available information as described by Levine-Schnur (2023). If the purpose of taking was not solely for general “traditional” or “pure” public use, i.e., collective goods, it was categorized as “Urban Renewal” (Hoyman and McCall 2010). All projects that were purposed for blight or economic development including both what could be labeled as slum remediation, blight removal, and economic development projects where the property taken cannot be said to inflict harm on society were marked as “Urban Renewal.”

Figure 1 shows the annual count of projects and affected BBLs by broad classification and year. As evidenced in the top panel, urban renewal projects constituted a large share of all projects in the early-to-mid 1990s before their use began to slow. The most recent urban renewal projects occurred in 2009. The bottom panel in Figure 1 shows the number of unique BBLs that were directly involved in an urban renewal or non-urban renewal project each year between 1990 and

2019.

[Figure 1 here]

3.2 Economic Data

The theoretical framework developed in Section 2.2 yields predictions about three classes of outcomes: employment at existing firms (the intensive margin), the count of active establishments (the extensive margin), and assessed property values (the capitalization channel). We construct measures of each at the BBL level from separate administrative and commercial data sources.

Employment and establishment counts are drawn from the Your Economy Time Series (YTS) database, a partnership between the private-sector marketing firm Data Axle and the University of Wisconsin Business Dynamics Research Consortium (UWBDRC). First publicly released in 2016, the YTS database is designed to capture the universe of business establishments operating in the United States and can best be described as a private-sector alternative to the Census Bureau’s restricted-access Longitudinal Business Database. The database begins in 1997 and includes almost 36 million distinct establishments through 2024. Major data fields include annual employment counts, latitude and longitude coordinates, primary North American Industrial Classification System (NAICS) and Standard Industrial Classification (SIC) codes, and entry and exit dates. Additional fields include linkages to parent or headquarter companies, Security and Exchange Commission central index key (CIK) and Committee on Uniform Securities Identification Procedures (CUSIP) identifiers for publicly traded firms, and indicators for veteran or minority-owned establishments (Miller et al. 2025). We use these data to construct BBL-level annual employment totals, which correspond to the intensive margin prediction of the model, and BBL-level annual establishment counts, which correspond to the extensive margin prediction.⁵

Property value outcomes are drawn from MapPLUTO, an annually updated parcel-level dataset

⁵For takings that result in private housing development, employment and establishment counts may be less natural benchmarks than for commercial or mixed-use projects, since residential development does not necessarily attract new businesses or generate persistent parcel-level employment. Residential redevelopment can nonetheless generate construction employment during build-out and some downstream demand for local retail and services, so these counts are not entirely uninformative for housing projects; whether effects of this kind are large or persistent enough to register at the parcel level given our sample sizes is an open question. The assessed land and building value outcomes provide a benchmark that applies to housing projects as well as commercial ones: if a taking generates genuine economic value for the surrounding neighborhood, that value should capitalize into nearby property prices regardless of project type.

published by the New York City Department of City Planning. MapPLUTO merges tax lot records from the Department of Finance with land use, zoning, and building characteristics from the Department of Buildings, and covers all five boroughs. Each record corresponds to a unique BBL and includes the total assessed value of the parcel as determined by the Department of Finance for property tax purposes, decomposed into its two components: assessed land value and assessed total building value. Because assessed values in New York City are updated annually and are anchored to the same administrative parcel identifiers used throughout our analysis, they provide a consistent and directly comparable measure of property value at the BBL level. The MapPLUTO data are available beginning in 2002, and we use the full 2002 to 2024 panel for our property value analysis. Assessed values for land and buildings are reported in millions of dollars. We analyze the two components separately because they capture distinct margins of the capitalization channel: land value reflects the market’s assessment of locational amenities and development potential, which should respond most directly to any amenity generated by a nearby taking, while building value additionally incorporates the existing capital stock and is therefore more likely to reflect prior investment decisions rather than contemporaneous neighborhood effects. These outcomes correspond to the capitalization prediction of the model: if a taking generates a positive amenity, that amenity should capitalize into assessed land and building values at nearby parcels, with the effect attenuating with distance.

We also examine the effect of takings on building permit activity, which is not a direct prediction of the theoretical framework but provides a useful window onto a distinct mechanism through which condemnation may affect surrounding parcels. Permit filings measure the flow of *intended* investment into a parcel, capturing decisions by property owners and developers to undertake new construction, alterations, or other improvements. If a taking raises the expected return to investment at nearby parcels, by raising land values or improving access to complementary uses, owners may respond by upgrading their properties, and this response would appear in permit activity before it is fully reflected in employment or assessed values. Building permit data therefore provide an early indicator of whether nearby landowners perceive the taking as generating positive spillovers, independently of whether those spillovers ultimately materialize in labor market outcomes.

Our building permit data come from the Department of Buildings (DOB) Permit Issuance dataset, which provides comprehensive records of all building permits issued by the NYC Department

of Buildings. This dataset captures permits for new construction, alterations, demolitions, and other building-related activities across all five boroughs. Each permit record includes the permit type, work description, filing date, issuance date, job start date, job cost, and precise geographic coordinates that allow us to link permit activity to specific parcels. These data are available as far back as 1989; however, we limit the sample to 1997 to 2024 to align with the employment and establishment data. One limitation is that the data track permit issuances but do not include completion information, meaning we can observe when construction activity is authorized to begin but not whether a project is actually undertaken or completed, which is why we use the term *intended* investment.

All five outcome variables exhibit over-dispersion, and we therefore transform each prior to estimation using the inverse hyperbolic sine, $\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$. We use this transformation rather than the natural logarithm because the log transformation requires strictly positive values, which would force us to exclude BBLs with zero outcomes from the analysis.

3.3 Treatment and Comparison Groups

We use establishment-level latitude and longitude coordinates to construct economic outcome measures at the BBL (parcel) level. To identify causal effects of taking projects, we compare economic outcomes in BBLs directly affected by takings, along with nearby BBLs, to BBLs that have never been involved in, or located within 200 meters of, any condemnation site since 1990. We estimate multiple specifications as robustness checks to account for potential spillover effects from project sites to BBLs that are nearby but not directly involved.

The selection of these treatment groups is motivated by the decay function $g(d)$ from our theoretical framework and corroborated by existing empirical evidence on the spatial reach of agglomeration spillovers. The monocentric model’s core mechanism is proximity as cost reduction: if the taking creates a complementary facility, nearby parcels benefit through reduced customer access costs, shorter worker commutes, and tighter supply chain linkages, with these advantages eroding with distance.⁶

⁶The cost-reduction mechanism maps most directly onto commercial redevelopment, where a nearby establishment or retail cluster lowers transport and input costs for surrounding firms. Residential redevelopment operates through parallel channels. A larger resident population raises local demand for retail and services, increasing the consumer amenities accessible from nearby parcels, and new construction generally replaces vacant or distressed structures, removing a local disamenity. Each of these effects capitalizes into surrounding land prices and diminishes with distance, so the distance-attenuation prediction holds whether the redeveloped use is commercial or residential.

Previous studies using geocoded establishment-level microdata consistently find that spillovers attenuate rapidly and within narrow radii. Arzaghi and Henderson (2008), studying advertising agencies in Manhattan, find that networking spillovers lose approximately 80 percent of their value within 500 meters and dissipate entirely by 750 meters. Ahlfeldt et al. (2015), using the division and reunification of Berlin as a natural experiment, estimate production externalities that reach roughly zero within half a mile. The bulk of this literature, however, operates at geographic scales considerably larger than a single parcel, suggesting that 200 meters represents a reasonable outer bound for detecting spillovers at the parcel level. We construct four distinct treatment groups (T1 through T4) to trace the spatial gradient within this bound: from the directly condemned parcel (T1), to adjacent BBLs (T2), to all parcels within 100 meters (T3), and finally to all parcels within 200 meters (T4). Comparing estimates across these groups allows us to test directly whether any effects decay with distance from the taking site, as the theory predicts.

[Table 1 here]

Table 1 outlines four treatment groups and four comparison groups of BBLs. We estimate 16 specifications for each economic outcome by combining every treatment group with every comparison group (4×4 combinations). Figure 2 provides a concrete illustration of how these groups are constructed, showing the treatment and baseline comparison group, from one taken parcel of the (much larger) Atlantic Yards project that occurred in Brooklyn.

[Figure 2 here]

Our baseline specification defines treatment group T1 as comprising exclusively the BBLs that are directly involved in a project (urban renewal vs. non-urban renewal). In Figure 2, this individual project BBL is depicted in dark purple shading. Our empirical approach compares economic outcomes in these treatment BBLs across the pre- and post-project periods against four distinct comparison groups, each defined using different proximity criteria to test the robustness of our findings.

The first comparison group, C1, encompasses all BBLs situated between 200 and 400 meters from any BBL directly involved in a condemnation project. These comparison BBLs appear as

green-shaded areas in Figure 2. We apply a strict exclusion criterion to all comparison groups: any potential comparison BBL that falls within 200 meters of any condemnation project (whether urban renewal or alternative public purpose) since 1990 is removed to ensure these units remain truly unaffected by taking activities.

Recognizing that condemnation projects may generate spillover effects on nearby parcels, we construct three additional treatment groups with progressively broader geographic scope. Treatment group T2 expands the definition to include both the directly involved project BBLs and all immediately adjacent BBLs, with the adjacent parcels shown in magenta shading in Figure 2. Treatment group T3 further broadens this definition to encompass all BBLs located within 100 meters of any project BBL, represented by dark pink shading in the figure. Treatment group T4 includes all BBLs within 200 meters of a project BBL, shown in light pink shading.

To complement these varied treatment definitions, we construct three additional comparison groups using different distance ranges. Comparison group C2 includes BBLs located between 200 and 600 meters from project sites. Comparison groups C3 and C4 employ distance thresholds of 400 to 600 meters and 400 to 800 meters, respectively, deliberately excluding the closer comparison BBLs that appear in C1 and C2 to create an additional buffer between the treatment site and the donor pool.

The four comparison groups serve two distinct purposes. First, each treatment-comparison pairing is designed to estimate the ATT using a donor pool whose members share similar locational characteristics with the treated BBLs by virtue of geographic proximity, while remaining far enough from the project site to be unaffected by the taking itself. The inner boundary of each comparison group is set at 200 meters, which is also the outer boundary of the broadest treatment group (T4), ensuring that no BBL can simultaneously serve as a treated and a comparison unit. Second, because the precise distance at which any amenity spillover dissipates to zero is unknown, the four comparison groups allow us to assess whether results are sensitive to where that threshold is assumed to lie. C1 and C2 cast a wider net by including BBLs relatively close to the project site, while C3 and C4 impose a more conservative buffer by excluding the 200 to 400 meter ring entirely. If estimates are stable across C1 through C4, we can be confident that our results are not driven by the choice of comparison group or by spillovers contaminating the donor pool at the inner boundary.

[Table 2 here]

Depending on treatment group definitions (T1 through T4), the employment, establishment, and permit analysis examines between 122 and 2,787 BBLs affected by urban renewal condemnations and between 1,906 and 23,045 BBLs affected by non-urban renewal condemnations. Our identification strategy requires at least two pre-treatment observations for each BBL. For employment, establishment, and permit outcomes, this restricts the sample to condemnations occurring in 1999 or later. For property value outcomes, the later start date of the MapPLUTO data requires condemnations to occur in 2004 or later. The property value analysis therefore covers a subset of these BBLs, ranging from 88 to 1,679 for urban renewal projects and from 1,609 to 19,332 for non-urban renewal projects.

Table 2 presents descriptive statistics for all outcome variables across treatment and comparison groups, stratified by project type. Employment, establishment, and building permit data cover the full sample period from 1997 to 2024; assessed land and building values cover 2002 to 2024 and are reported in millions of dollars. For urban renewal projects, the narrowest specification (T1 and C1) shows treated BBLs with substantially higher baseline activity: average employment of 14.2 compared to 5.9 in comparison areas, nearly double the number of business establishments (0.98 vs. 0.49), and more than double the building permit issuances (0.63 vs. 0.29). Assessed land and building values follow the same pattern, with treated BBLs exhibiting markedly higher values than comparison BBLs. As the geographic definitions broaden to T4 and C4, these differences moderate across all outcomes. The large standard deviations reveal considerable within-group heterogeneity, particularly for employment levels and assessed values. For non-urban renewal projects, the pattern reverses at the narrowest specification: comparison BBLs average 9.2 employees compared to 3.0 in treatment areas, and maintain higher establishment counts and assessed values, suggesting that non-urban renewal takings tend to occur in areas with lower baseline economic activity relative to their surroundings.

3.4 Identification Strategy

With our sample of treated BBLs (those involved in condemnation projects) and comparison BBLs (those never involved in or near condemnation projects), a regression-based difference-in-differences approach offers a straightforward method for estimating the average treatment effect

on treated BBLs. However, causal identification through difference-in-differences relies on the assumption that comparison group trends represent a valid counterfactual for the treated group. This parallel trends assumption is satisfied when comparison BBLs accurately reflect the trajectory that treated BBL outcomes would have followed in the absence of condemnation activity.

The parallel trends assumption cannot be verified directly, since it concerns unobserved counterfactual outcomes. However, it is standard to examine whether treated and comparison groups follow similar trajectories in the pre-treatment period as an indirect test of the assumption’s plausibility. Parallel pre-treatment trends provide reassurance that the assumption may continue to hold after treatment begins.

In our data, we found statistically significant differences in pre-treatment trends between treated and comparison BBLs, which undermines confidence in the parallel trends assumption. To address these identification concerns and produce more credible causal estimates, we turn to the partially pooled synthetic control (PPSC) methodology of Ben-Michael et al. (2022) for estimating the average treatment effect on treated BBLs.⁷

The level differences visible in Table 2 do not themselves pose an identification problem. Synthetic control does not require treated and comparison BBLs to have similar average levels of economic activity. It requires only that a weighted combination of comparison BBLs reproduce the pre-treatment trajectory of the treated BBLs.

These level differences reflect the fact that condemned parcels, particularly those in urban renewal projects, tend to sit in denser, more commercially active areas than the surrounding comparison parcels (Levine-Schnur 2023). The partially pooled synthetic control absorbs them through unit-level intercept shifts, which remove time-invariant level gaps and let the estimator match pre-treatment dynamics rather than levels. The counterfactual can therefore follow a treated BBL even

⁷Two prominent alternative estimators merit brief discussion. The matrix completion (MC) method of Athey et al. (2021) imposes a low-rank factor structure on the untreated potential outcomes panel and recovers missing counterfactual entries via nuclear norm minimization. The MC approach is well suited to settings where outcomes follow a latent factor structure and the missing data pattern is relatively simple. However, the staggered adoption setting with many treated cohorts that characterizes our data does not map cleanly onto the standard MC framework without additional assumptions about how the low-rank structure interacts with treatment timing. The augmented synthetic control method of Ben-Michael et al. (2021) adds a bias correction term to standard SCM estimates using an outcome model fit on the donor pool, and is most naturally suited to settings with a single treated unit or a small number of treated units estimated separately. The PPSC framework of Ben-Michael et al. (2022) was developed specifically for settings with many treated units and staggered adoption timing, and has been shown to dominate both fully separate and fully pooled SCM in terms of finite sample estimation error bounds. These properties align directly with the structure of our data, and we therefore treat the PPSC estimates as our primary results.

when its baseline activity differs substantially from any individual comparison BBL.

This also resolves the pre-treatment trend difference noted earlier. That difference concerns treated BBLs and the full pool of nearby comparison BBLs, and it is what moves us away from difference-in-differences. Synthetic control does not depend on that raw comparison; identification turns instead on close pre-treatment agreement between each treated unit and its synthetic counterfactual, not between the treated and comparison pools as a whole.

The event study plots in Section ?? (Figure ??) show that this condition holds: treated units and their synthetic controls track closely throughout the pre-treatment window for every outcome and both project types.

The PPSC is designed to handle settings with multiple treated units, potentially heterogeneous treatment effects, and staggered adoption. Their framework starts with the fundamental synthetic control problem: for treated unit j , find weights $\phi_j = (\phi_{1j}, \phi_{2j}, \dots, \phi_{Nj})$ that solve:

$$\min_{\phi_j \in \Phi_j} \frac{1}{L_j} \sum_{k=1}^{L_j} \left(Y_{j,T_j-k} - \sum_{i=1}^N \phi_{ij} Y_{i,T_j-k} \right)^2 + \lambda \sum_{i=1}^N \phi_{ij}^2, \quad (11)$$

where Φ_j represents the probability simplex and L_j is the number of available pre-treatment observations for unit j . This approach improves upon traditional synthetic control methods by directly targeting prediction accuracy in the pre-treatment period (Doudchenko and Imbens 2016) while incorporating regularization to prevent overfitting (Ben-Michael et al. 2022).

The optimization problem in equation (11) represents a departure from the original Abadie (2005) and Abadie et al. (2010) framework in two important respects. Rather than employing a nested optimization procedure that first matches on auxiliary covariates, this specification directly minimizes pre-treatment prediction error. Second, the inclusion of the ℓ_2 penalty term $\lambda \sum_{i=1}^N \phi_{ij}^2$ helps mitigate overfitting concerns that can arise when the number of potential donor units is large relative to the available pre-treatment periods (Abadie et al. 2015).

Using the optimal weights $\hat{\phi}_j$ from equation (11), the synthetic control estimate of the counterfactual outcome for treated unit j in post-treatment period k is constructed as:

$$\hat{Y}_{j,T_j+k} = \sum_{i=1}^N \hat{\phi}_{ij} Y_{i,T_j+k}, \quad (12)$$

with the corresponding treatment effect estimate given by $\hat{\tau}_{jk} = Y_{j,T_j+k} - \hat{Y}_{j,T_j+k}$.

The quality of the synthetic control can be assessed through the pre-treatment fit measure, which computes the root mean squared prediction error over the pre-treatment period. When applied to pre-treatment data (i.e., $k < 0$), the difference between observed and synthetic outcomes represents a placebo treatment effect. Setting the regularization parameter to zero, this placebo effect can be expressed as the mean squared error of pre-treatment predictions (Ben-Michael et al. 2022).⁸

Formally, this is given by:

$$\left(q_j(\hat{\phi}_j)\right)^2 = \frac{1}{L_j} \sum_{k=1}^{L_j} \left(Y_{j,T_j-k} - \sum_{i=1}^N \hat{\phi}_{ij} Y_{i,T_j-k}\right)^2. \quad (13)$$

Perfect pre-treatment fit occurs when $q_j(\hat{\phi}_j) = 0$, indicating that the synthetic control exactly replicates the treated unit's pre-treatment trajectory.

The key innovation of Ben-Michael et al. (ibid.) is to extend this single-unit framework to settings with many treated units and staggered timing. With many treated BBLs rather than one, a choice arises. One option is to build a separate synthetic control for each treated BBL, matching each unit's own pre-treatment path as closely as possible. The other is to pool the treated BBLs and build a single synthetic control for the average treated unit. Fitting each unit separately tracks individual histories well but can be noisy; pooling is more stable but can fit any single unit poorly. Their partially pooled SCM balances these two objectives, borrowing strength across units through pooling while preserving enough unit-specific fit to keep the individual counterfactuals credible.

When there are J treated units, Ben-Michael et al. (ibid.) show that the average treatment

⁸Ben-Michael et al. (2022) demonstrate that λ becomes more important when the ratio of treated units to pre-treatment periods increases. In our application, varying λ has minimal impact on the estimated weights, so we set $\lambda = 0$ throughout our analysis. This insensitivity is expected in our setting because the ridge penalty in equation (12) penalizes the Frobenius norm $\|\Phi\|_F^2$, shrinking weights toward uniformity across the donor pool. A lasso-type penalty would instead induce sparsity, concentrating weight on a smaller number of donor BBLs. In our setting, however, sparsity is not obviously desirable because the donor pool consists of nearby parcels selected on geographic proximity, and distributing weight across this pool is consistent with constructing a valid local counterfactual. Moreover, the partially pooled objective already regularizes estimation by jointly optimizing pooled and unit-specific fit across cohorts, reducing the marginal contribution of any additional penalty. Doudchenko and Imbens (2016) propose an elastic net formulation that nests both ridge and lasso as special cases; under their framework, the near-zero sensitivity to λ we observe supports the ridge-dominant specification we employ.

effect on the treated in period k admits two equivalent representations:

$$\widehat{ATT}_k = \frac{1}{J} \sum_{j=1}^J \hat{\tau}_{jk} = \underbrace{\frac{1}{J} \sum_{j=1}^J \left[Y_{j,T_j+k} - \sum_{i=1}^N \hat{\phi}_{ij} Y_{i,T_j+k} \right]}_{\text{average of separate synthetic controls}} = \underbrace{\frac{1}{J} \sum_{j=1}^J Y_{j,T_j+k} - \sum_{i=1}^N \sum_{j=1}^J \frac{\hat{\phi}_{ij}}{J} Y_{i,T_j+k}}_{\text{pooled synthetic control approach}} \quad (14)$$

This equivalence reveals that average treatment effects can be computed either by averaging unit-specific estimates or by constructing a synthetic control for the average treated unit. Under common data-generating processes such as autoregressive models or linear factor structures, the estimation error in $\widehat{ATT}_k$ decomposes into components reflecting individual unit fit quality, pooled fit quality, and random noise (Ben-Michael et al. 2022).

Building on the relationship in equation (14), Ben-Michael et al. (ibid.) develop their partially pooled estimator by minimizing a weighted combination of pooled and separate pre-treatment prediction errors. Let $\alpha \in \mathbb{R}^J$ represent potential intercept differences between treated and control units, and let $\Phi = [\phi_1, \phi_2, \dots, \phi_J]$ denote the $N \times J$ matrix collecting all weight vectors. The partially pooled SCM solves:

$$\min_{\alpha \in \mathbb{R}^J, \Phi \in \Phi_{scm}} \nu (\tilde{q}^{pool}(\alpha, \Phi))^2 + (1 - \nu) (\tilde{q}^{sep}(\alpha, \Phi))^2 + \lambda \|\Phi\|_F^2, \quad (15)$$

where $\tilde{q}^{pool}$ and $\tilde{q}^{sep}$ represent normalized measures of pooled and separate pre-treatment prediction errors, respectively.⁹ The Frobenius norm $\|\Phi\|_F^2$ penalizes the sum of squared weights across all units.

The hyperparameter $\nu \in [0, 1]$ governs the relative emphasis placed on pooled versus separate estimation approaches. At the extremes, $\nu = 0$ corresponds to estimating individual synthetic controls for each treated unit while ignoring aggregate fit, whereas $\nu = 1$ prioritizes the synthetic control for the average treated unit while disregarding individual unit fits. Ben-Michael et al. (ibid.) propose a data-driven procedure for selecting ν that balances these competing objectives based on the relative quality of pooled and separate fits. Our estimates are based on the value of ν selected from their data-driven heuristic.¹⁰

⁹Complete derivations of these normalized error expressions appear in Ben-Michael et al. (2022).

¹⁰We also re-estimated our specifications setting $\nu = 1$, which focuses the optimization entirely on minimizing the prediction error for the overall average treatment effect. There are no meaningful differences using this alternative approach.

As Ben-Michael et al. (2022) demonstrate, the partially pooled approach can incorporate unit-level intercepts and allow for pre-treatment matching on additional covariates. Inference is performed using a wild bootstrap procedure adapted from Otsu and Rai (2017). The average treatment effect is first rewritten as a weighted sum over all units (both treated and control), where each unit contributes a specific term to the overall estimate. For each bootstrap iteration, random multiplier variables are drawn from a two-point distribution designed to preserve the variance structure of the original estimator. With the estimated synthetic control weights and outcome data held fixed across bootstrap replications, the resampled treatment effect statistic is computed by multiplying each unit’s contribution by its corresponding random multiplier and summing across all units. Confidence intervals are then constructed using the quantiles of the resulting bootstrap distribution of treatment effect estimates.

One inferential consideration specific to our setting concerns the multi-parcel structure of eminent domain projects. Treatment is assigned at the project level, and a single project may affect multiple parcels whose economic outcomes may be correlated. Drawing bootstrap multipliers $W_i^{(b)}$ at the parcel level would implicitly treat parcels as independent units of randomization, producing confidence intervals that are too narrow whenever within-project correlation is present. This, in turn, would bias inference toward finding statistically significant effects. We therefore draw bootstrap multipliers at the project level. The distribution of parcels per project in our raw takings data is highly right-skewed: the median project affects 3 parcels, 34% of projects affect a single parcel, and 73% affect 10 or fewer. A small number of large infrastructure projects account for the remainder. Bootstrapping at the project dimension accounts for within-project correlation across economic outcomes and should yield more conservative confidence intervals.

4 Results

4.1 Baseline Results

With four treatment groups and four comparison groups, we estimate a total of 16 specifications for each of the five economic outcomes: employment, business establishments, building permit issuances, assessed land value, and assessed building value. Because all outcome variables exhibit over-dispersion (variance > mean), we transform each outcome using the inverse hyperbolic sine (ihs) to reduce dispersion prior to estimation. We use this transformation rather than the natural

logarithm because the log transformation requires strictly positive values, which would force us to exclude BBLs with zero outcomes from the analysis.¹¹

As noted in Section 3.4, the partially pooled synthetic control approach requires a minimum of two pre-treatment observations for each treated BBL. Given the staggered timing of condemnation projects, the synthetic counterfactuals are formed using pre-treatment data ranging from a minimum of two years prior to treatment up to a maximum of twelve years. We also limit the post-treatment period to 10 years because, over longer time horizons, it becomes increasingly difficult to attribute observed changes to the original condemnation rather than other intervening factors. Economic cycles, policy changes, and demographic shifts can all influence property values and neighborhood dynamics, making it harder to isolate the condemnation’s specific effects after a decade.

Estimates of the ATT for all specifications involving urban renewal projects are presented in Figure 3. Each panel represents a different outcome variable, with every panel summarizing estimates from all 16 specifications. Different treatment groups are identified by distinct line markers, and error bars show 95% confidence intervals. For example, the point estimate for urban renewal projects using Treatment Group 4 and Comparison Group 1 appears as T4-C1 in Figure 3, identified by a green triangle.

Comparing estimates across the four comparison groups for any given treatment group provides a direct test of whether the results are sensitive to the assumed location of the attenuation threshold: if the estimated ATT is stable across C1 through C4, this supports the conclusion that the donor pool is not contaminated by spillovers at its inner boundary and that the choice of comparison group does not drive the findings.

[Figure 3 here]

Focusing on the far left panel in Figure 3, the employment point estimates are negative across

¹¹The inverse hyperbolic sine transformation, $\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$, offers significant advantages over the commonly used $\ln(x + 1)$ adjustment for handling zero values. The traditional approach of adding one before taking the logarithm creates interpretational challenges and introduces bias for small values of x due to the discontinuous behavior at zero. The inverse hyperbolic sine avoids these problems by providing a smooth, continuously differentiable function that behaves similarly to the logarithm for large values while remaining well-defined for all real numbers, including zeros and negatives (Bellemare and Wichman 2020). This transformation also naturally addresses over-dispersion by compressing extreme values, thereby reducing variance in the outcome variable. These properties make it particularly suitable for estimating elasticities and semi-elasticities with improved precision.

13 of 16 specifications, with a median ATT of -0.04 , corresponding to roughly a 4% reduction relative to the synthetic counterfactual under the inverse hyperbolic sine approximation.¹² None of the 16 estimates achieves statistical significance at conventional levels. It is worth noting that point estimates are largest in magnitude for T1, the directly condemned parcels, with ATTs around -0.17 , and attenuate steadily as the treatment definition broadens through T2 and into T3 and T4. The T1 confidence intervals are wide given the small number of directly condemned urban renewal BBLs (122), and the broader treatment groups yield estimates that are precisely centered near zero. Taken together, the employment evidence offers no support for the view that urban renewal condemnations generate positive labor market effects; if anything, the preponderance of negative point estimates runs in the opposite direction from what the economic development rationale would predict.

The middle panel presents results for business establishments. Fourteen of 16 point estimates are negative, with a median ATT of -0.02 , and a similar attenuation pattern from T1 through T4 is visible here as well. The T2 specifications produce the estimates closest to conventional significance thresholds: the T2-C1 ATT of -0.042 carries a t -statistic of -2.10 , and the remaining T2 specifications yield t -statistics between -1.43 and -1.81 . As with employment, the direction of the evidence is inconsistent with positive economic development effects. The T2 results suggest, if anything, a modest negative effect on establishment counts in parcels adjacent to condemned lots, though the evidence does not establish this firmly across the full set of specifications.

The right panel examines building permits and shows a different and more consistent pattern. All 16 point estimates are positive, and 12 of the 16 are statistically significant at conventional levels, with significant ATTs ranging from approximately $+0.017$ to $+0.088$. The significant results are concentrated in treatment groups T2, T3, and T4; T1 estimates are also positive but imprecisely estimated, consistent with limited statistical power in the smallest treatment group. These permit results stand out as the only outcome for urban renewal projects where the evidence is consistently positive and statistically significant across a majority of specifications. Two features of the permit data limit how much weight this finding can bear, however. First, the absolute magnitudes are modest: the mean number of permits issued in the treatment group is 0.628 (Table 2), and even

¹²For variables transformed using the inverse hyperbolic sine, coefficients can be interpreted approximately as percentage changes when the values are not close to zero, similar to logarithmic transformations (Bellemare and Wichman 2020). The actual coefficient estimate can be recovered using the hyperbolic sine ($\sinh(x) = \frac{e^x - e^{-x}}{2}$).

the largest significant ATT of +0.088 implies an increase of only around 0.055 permits issued per year relative to the counterfactual. Second, and more fundamentally, issued permits reflect the city’s approval of a proposed work plan rather than the execution of that plan. Unlike employment, establishment counts, or assessed property values, a permit issuance confirms only that construction has been authorized to begin, not that it was actually undertaken or completed. We have no data on the fraction of issued permits that ultimately result in completed projects, so the permit results cannot be straightforwardly converted into a claim about actual economic output.

The property value panels complete the urban renewal results, and both are uniformly negative in sign. For assessed land value, all T4 specifications are statistically significant, with ATTs ranging from -0.085 to -0.098 (t -statistics between -2.58 and -3.17), and the T2 specifications are consistently negative but just below conventional thresholds. For T1, point estimates are positive but extremely imprecise (standard errors around 0.50), and little weight should be placed on their sign given the small sample. For assessed building value, all 16 point estimates are negative and the T4 specifications are statistically significant (ATTs from -0.166 to -0.190).

Across all 80 urban renewal specifications, 55 point estimates are negative. The 25 positive estimates are concentrated entirely in the building permits outcome, which, as noted above, reflects the city’s approval of proposed work plans rather than realized economic activity. While the theoretical framework can plausibly be extended to predict increased permit issuances near a successful taking site, the 12 statistically significant positive permit estimates stand against 9 statistically significant negative estimates spread across employment, establishments, and property values, which are the outcomes most closely associated with the economic development rationale for condemnation. This overall pattern is difficult to reconcile with the claim that urban renewal takings generate positive economic spillovers.

[Figure 4 here]

We also provide separate estimates for non-urban renewal condemnations, shown in Figure 4. These projects encompass takings for traditional public purposes such as infrastructure development, schools, parks, and other government facilities. The non-urban renewal sample is considerably larger than the urban renewal sample, with between 1,906 and 23,045 treated BBLs depending on the treatment group definition, and the estimates are correspondingly more precise.

The results across employment and business establishments demonstrate a consistent pattern. For employment, all 16 point estimates are negative, with a median ATT of -0.01 , and the T2 specifications reach conventional significance thresholds across multiple comparison groups (t -statistics from -1.88 to -2.86 , ATTs around -0.015 to -0.020). The T1 estimates approach but do not cross conventional thresholds (t -statistics between -1.45 and -1.91), and point estimates attenuate from T1 through T4. The business establishment panel closely parallels the employment findings: 14 of 16 point estimates are negative, and T2 specifications again produce the most precisely estimated results, with t -statistics between -2.00 and -2.25 across comparison groups. As with employment, the point estimates are largest in magnitude for T1 and attenuate with distance. The direction of the evidence for both labor market outcomes is again the opposite of what the economic development rationale would predict.

For building permits, all 16 point estimates are positive and 5 specifications reach statistical significance, with significant ATTs in the range of $+0.004$ to $+0.005$. Under the inverse hyperbolic sine approximation, these estimates correspond to increases of less than 1% relative to the counterfactual, a magnitude that is economically modest regardless of which comparison group is used. As with the urban renewal results, these estimates reflect the city’s approval of proposed construction plans rather than confirmed completion of work.

Among non-urban renewal condemnations, the two property value outcomes diverge sharply. Assessed land value results are mixed across specifications, with only a small number reaching conventional significance and no clear directional pattern. Assessed building values, by contrast, are the most consistent results in the paper: all 16 point estimates are negative, all 16 are statistically significant, and t -statistics range from -3.66 to -12.50 in absolute value, with ATTs from -0.169 to -0.452 . This consistency across every specification, treatment group, and comparison group provides strong and consistent evidence of a negative effect on assessed building values near non-urban renewal project sites. Takings for schools, parks, transit infrastructure, and similar purposes typically remove existing buildings from the tax roll without generating the private redevelopment activity that an economic development condemnation might promise, and the consistent negative building value pattern is compatible with this mechanism. We note that these findings speak only to private property value losses in the vicinity of project sites; whether the broader social value of the public facility provided by the taking outweighs those private losses is a separate question that

lies outside the scope of this analysis.

4.2 Event Study Formulations

Figure 5 presents event study plots for each outcome using the T4-C4 specification, which pairs the largest treatment group with the most conservative comparison group. We show this pairing because it is the most demanding test of pre-treatment fit: T4 includes the most treated lots (all within 200 meters of a project site), providing the most statistical power, while C4 draws comparison lots from outside the full buffer zone, minimizing contamination concerns. Close fit here is therefore not an artifact of a small or favorably chosen sample. The full set of 160 event study plots (5 outcomes $\times$ 16 specifications $\times$ urban renewal/non-urban renewal) appears in Appendix A.¹³

These plots show the difference between observed outcomes in treated BBLs and their synthetic counterfactuals over time, with the vertical dashed line at time zero marking the beginning of the condemnation project. The pre-treatment period (left of the dashed line) reveals close fit between treated units and their synthetic controls across all outcomes for both urban renewal and non-urban renewal projects. The point estimates consistently hover around zero in the pre-treatment years, with confidence intervals that include zero throughout, indicating that our synthetic control methodology constructs counterfactuals that track the treated units' pre-intervention trajectories well.

[Figure 5 here]

The quality of pre-treatment fit shown in these event studies provides support for the parallel trends assumption that underlies our causal identification. Unlike traditional difference-in-differences approaches that assume parallel trends without the ability to construct explicit counterfactuals, the synthetic control method allows us to observe directly how well our comparison units replicate the treated units' pre-treatment behavior. This pre-treatment balance gives us confidence that post-treatment deviations can be attributed to the intervention rather than pre-existing differences between treated and comparison areas.

¹³As shown in Appendix A, the event study plots differ very little across treatment and comparison group specifications, and all synthetic counterfactuals have excellent pre-treatment fit with the outcome of interest. Much like the point estimates in Figures 3 and 4, the T4-C4 pairing shown here is therefore representative of the full set. Selecting a smaller treatment group for the main figure would have introduced concerns about cherry-picking a specification with unusually good pre-treatment fit.

It is worth clarifying the relationship between the event study plots in Figure 5 and the ATT estimates reported in Figures 3 and 4. The point estimates in those figures represent averages of the treatment effect over the full post-treatment window. The event study plots, by contrast, display the estimated effect at each individual year relative to treatment. A reader examining Figure 5 may notice that the estimated effect at a particular post-treatment year occasionally diverges from zero in a way that appears statistically meaningful, even for outcomes where the overall ATT does not. This does not reflect an inconsistency in our results: a single elevated year within a ten-year post-treatment horizon does not imply a significant average effect, and it is the latter that our main figures report.

With this in mind, two of the six panels in Figure 5 show post-treatment trajectories that are broadly consistent with the statistically significant findings discussed above: the urban renewal building permits panel shows a positive divergence in the later post-treatment years, and the non-urban renewal building value panel shows a sustained negative divergence across much of the post-treatment horizon. For employment and establishments, the event study plots show no systematic post-treatment deviation for either project type, consistent with the absence of statistically detectable effects for those outcomes.

4.3 Discussion

The results across both urban renewal and non-urban renewal condemnations do not provide strong support for the view that eminent domain for economic development generates positive economic spillovers. Across all 160 specifications, 80 for urban renewal and 80 for non-urban renewal, 111 point estimates are negative (69.3% of all estimates). The 49 positive estimates are concentrated almost entirely in the building permits outcome, which reflects the city’s approval of proposed construction plans rather than realized economic activity: unlike employment, establishment counts, or assessed property values, a permit issuance confirms only that construction has been authorized to begin, not that it was actually undertaken or completed. Of the 17 statistically significant positive estimates across all specifications (10.6% of all estimates), all are for building permits. By contrast, 33 specifications yield statistically significant negative estimates, spread across employment, establishments, and property values. While the theoretical framework can plausibly be extended to predict increased permit issuances near a successful taking site, the permit magnitudes

are modest relative to mean rates, and the absence of any corresponding positive signal in the outcomes that most directly reflect economic development (employment, business formation, and property values) leaves the overall evidence inconsistent with the economic development rationale underlying development-oriented condemnations.

Several complementary explanations may account for this pattern. Private markets may already coordinate land assembly reasonably well without government intervention, as professional real estate developers routinely navigate holdout problems through strategies such as option contracts, straw purchasers, and phased acquisition (Cunningham 2013). Recent experimental evidence supports this view: Isaac et al. (2016) find that competition among landowners can achieve aggregation rates exceeding 84% in competitive settings, while Portillo (2019) finds that private markets can capture approximately 65% of externality value through negotiation. In dense urban environments like New York City, the abundance of alternative development sites may further reduce the strategic value of any single condemned parcel.

A second explanation centers on displacement and crowding-out effects, where eminent domain may relocate rather than create net new economic activity. This interpretation is consistent with Chen and Yeh’s (2023) findings of greater displacement, particularly among racial minorities, in areas with more condemnations. If condemnation projects displace existing businesses and residents, new economic activity at project sites may be offset by losses elsewhere in the local economy. Our methodology examines effects in surrounding areas up to 800 meters from project sites, yet finds no consistent positive spillover effects on employment or establishments in any distance ring, suggesting that if condemnation projects do generate new activity, it does not produce broader neighborhood-wide gains that are detectable in these data.

Several limitations of the analysis deserve acknowledgment. First, our findings are representative only of New York City, and some modesty about external validity is warranted: eminent domain processes vary across jurisdictions in terms of political oversight, planning capacity, and project selection, and the institutional features that make New York City an attractive setting for this analysis, particularly the regulatory stability that followed *Kelo*, may not generalize to other contexts. Second, we have not examined the specific economic benefits identified by the condemning authority in the Determinations and Findings for each project, and it is possible that some projects were justified on grounds not fully captured by our five outcome measures. That

said, employment, business establishment counts, building permit issuances, and assessed land and building values collectively span the range of outcomes most commonly invoked to justify economic development takings, and the consistency of the evidence across all five provides a broad picture of the overall economic effects that is unlikely to be overturned by attention to any single project’s stated objectives.

Finally, the political economy of eminent domain may also contribute to the pattern of results. Local officials frequently work in close coordination with private developers on projects that utilize eminent domain, and the interests of those developers do not necessarily align with the broader goal of generating community-wide economic benefits. Site selection may reflect the preferences of a developer who has already identified a parcel rather than an independent government assessment of where investment would generate the greatest public return (U.S. Commission on Civil Rights 2014). When legal requirements mandate demonstrating public benefit, officials may develop post hoc economic justifications for projects whose foundations lie elsewhere (Peters and Fisher 2004). This interpretation is consistent with our findings: the absence of detectable spillover benefits to the surrounding community does not preclude private gains to the developer, and indeed both are consistent with a transfer story in which condemnation redistributes value from displaced property owners and surrounding parcels to private interests rather than generating net new economic activity. We are not aware of direct empirical evidence quantifying developer returns from condemnation projects, but to the extent that there is continued demand for eminent domain among large development interests this suggests that the practice delivers private value even where community-wide benefits are absent.

5 Conclusion

Our study provides a rigorous empirical test of the spillover benefit rationale commonly invoked to justify development-oriented condemnations, using granular, location-specific data at the city level. The results offer limited empirical support for that rationale. Employment and business establishment counts show no evidence of a positive increase near condemned parcels, assessed property values at surrounding parcels tend to fall rather than rise, and the only consistently positive signal is confined to building permit issuances, an outcome that reflects the city’s approval of proposed construction plans rather than confirmed economic activity.

While our analysis cannot definitively rule out all potential economic impacts, the comprehensive methodology and robust statistical approach strongly suggest that takings for economic development do not generate positive economic benefits. These results raise fundamental questions about using eminent domain as an economic development tool. As the *Kelo* Court deferred to local governments on both the choice of public purpose and the efficacy of their development plans, our findings raise the question of whether policy-makers should subject their own economic projections to greater scrutiny before proceeding with a taking, and whether a more structured decision-making process might be warranted (see Levine-Schnur and Wagner 2025). Given the documented costs to displaced property owners and the absence of measurable community benefits, policymakers should reassess the presumed advantages of such land acquisition strategies and implement higher standards of economic justification when proposing takings.

Our findings offer empirical grounds for caution about the economic rationale underlying development-oriented takings, suggesting that the positive spillover effects commonly invoked to justify condemnation may not materialize as anticipated, at least in the New York City context studied here. Lopez et al. (2025) argue that post-*Kelo* reforms may lose effectiveness over time through “loophole mining” as developers and officials find creative ways around restrictions under diminished public scrutiny. Our results suggest that even in New York City, a jurisdiction that did not substantially reform its eminent domain laws following *Kelo*, the promised economic benefits fail to materialize. This finding strengthens the case for more stringent restrictions on economic development takings, as it demonstrates that the theoretical benefits used to justify such takings in practice do not occur, while the documented costs to displaced property owners remain very real.

The absence of measurable spillover effects in our comprehensive analysis suggests that local governments’ economic projections underlying taking decisions may be systematically overoptimistic. Combined with Lopez et al.’s (2025) evidence of ongoing regulatory circumvention in states that did implement reforms, our findings point toward a need for both stronger initial restrictions and ongoing vigilance against regulatory erosion over time.

Our analysis has limitations that future research should address. First, we cannot examine job quality or precise wage impacts, which would require even more granular employment data. Additionally, our study does not capture potential tax revenue implications or qualitative economic improvements that might not be reflected in employment and establishment metrics. At least in

the New York City context studied here, the evidence does not support the view that urban renewal takings deliver the economic transformations commonly invoked to justify them.

Future research could proceed along three complementary paths. First, researchers could develop methods to account for job quality and wage changes at the parcel level, investigate tax revenue implications, and explore whether qualitative economic improvements might offset the limited positive effects we document. Second, comparative cross-city studies could replicate this methodology in other urban contexts with different economic characteristics to assess whether our findings are New York City-specific or might extend to more diverse metropolitan settings. Third, and perhaps most relevant for policy, future work could develop frameworks for measuring the broader social benefits of takings for traditional public purposes, such as schools, parks, and transit infrastructure, and weigh those benefits against the private costs documented here. Our findings suggest that non-urban renewal takings are associated with measurable negative effects on surrounding property values, but whether those private losses are justified by the social value of the public facilities provided is a question that calls for a different empirical approach than the one employed here. These research directions could address some of the limitations of the current study and contribute to a more comprehensive understanding of the complex tradeoffs involved in the exercise of eminent domain power.

References

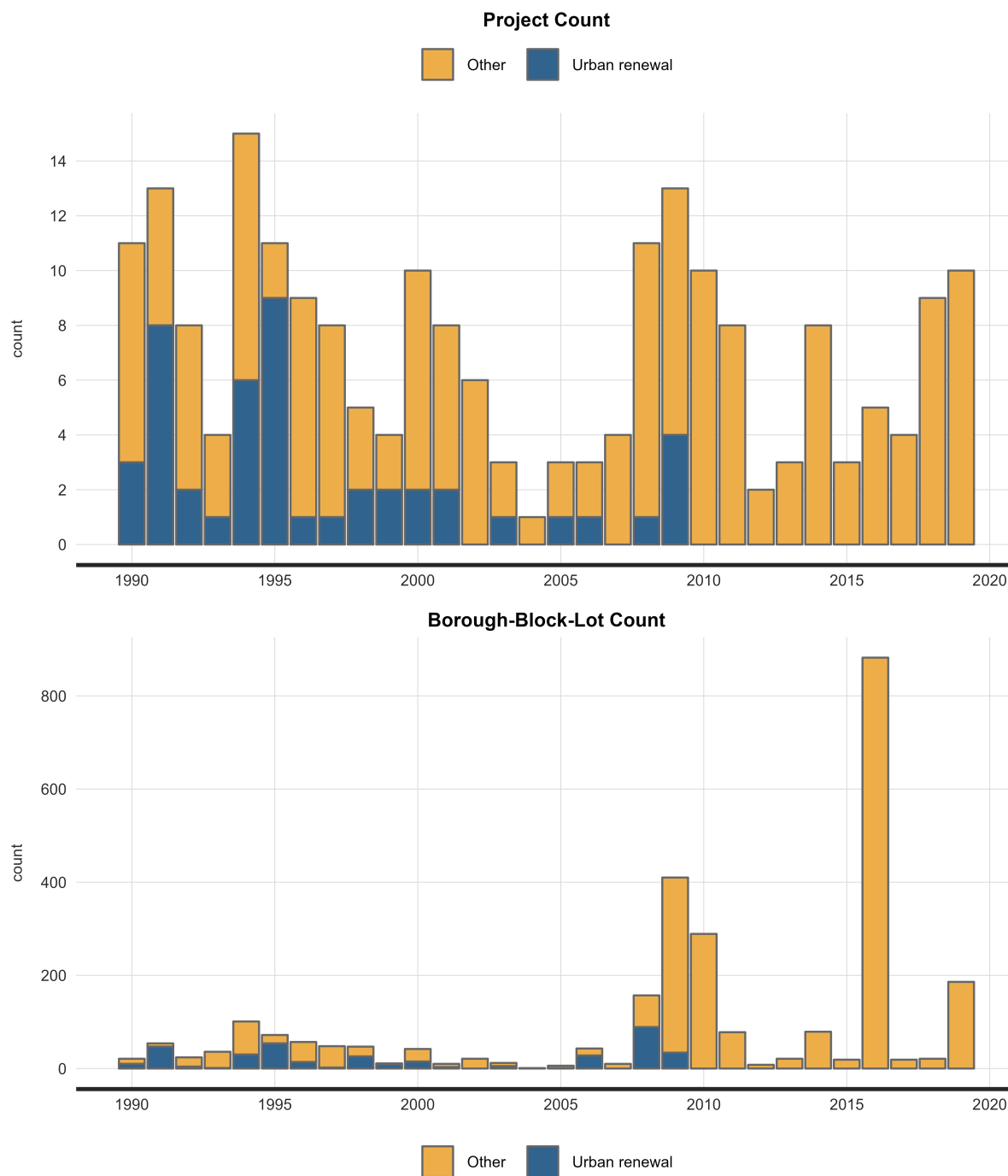
- Abadie, Alberto (2005). “Semiparametric Difference-in-Differences Estimators”. *Review of Economic Studies* 72, 1–19.
- Abadie, Alberto, Alexis Diamond and Jans Hainmueller (2010). “Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California’s Tobacco Control Program”. *Journal of the American statistical Association* 105 (490), 493–505.
- (2015). “Comparative Politics and the Synthetic Control Method”. *American Journal of Political Science* 59 (2), 495–510.
- Ahlfeldt, Gabriel M., Stephen J. Redding, Daniel M. Sturm and Nikolaus Wolf (2015). “The Economics of Density: Evidence from the Berlin Wall”. *Econometrica* 83 (6), 2127–2189. DOI: <https://doi.org/10.3982/ECTA10876>.
- Alonso, William (1964). *Location and Land Use*. Cambridge, MA: Harvard University Press.
- Arzaghi, Mohammad and J. Vernon Henderson (2008). “Networking off Madison Avenue”. *Review of Economic Studies* 75 (4), 1011–1038. DOI: <https://doi.org/10.1111/j.1467-937X.2008.00499.x>.
- Athey, Susan, Mohsen Bayati, Nikolay Doudchenko, Guido Imbens and Khashayar Khosravi (2021). “Matrix Completion Methods for Causal Panel Data Models”. *Journal of the American Statistical Association* 116 (536), 1716–1730.
- Bellemare, Marc F and Casey J Wichman (2020). “Elasticities and the inverse hyperbolic sine transformation”. *Oxford Bulletin of Economics and Statistics* 82 (1), 50–61.
- Ben-Michael, Eli, Avi Feller and Jesse Rothstein (2021). “The Augmented Synthetic Control Method”. *Journal of the American Statistical Association* 116 (536), 1789–1803. DOI: [10.1080/01621459.2021.1929245](https://doi.org/10.1080/01621459.2021.1929245). eprint: <https://doi.org/10.1080/01621459.2021.1929245>. URL: <https://doi.org/10.1080/01621459.2021.1929245>.
- (2022). “Synthetic controls with staggered adoption”. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 84 (2), 351–381. DOI: <https://doi.org/10.1111/rssb.12448>. URL: <https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/rssb.12448>.
- Berliner, Dana (2003). *Public Power, Private Gain: A Five-Year, State-By-State Report Examining the Abuses of Eminent Domain*. URL: https://ij.org/wp-content/uploads/2015/03/ED_report.pdf.
- (2006). *Opening the Floodgates: Eminent Domain Abuse in a Post Kelo World*. last visited Apr 15, 2020. URL: <https://ij.org/wp-content/uploads/2015/04/floodgates-report.pdf>.
- (2015). “Looking Back Ten Years after *Kelo*”. *Yale Law Journal* 125, 82.
- Blume, Lawrence, Daniel L Rubinfeld and Perry Shapiro (1984). “The Taking of Land: When Should Compensation be Paid?” *The Quarterly Journal of Economics* 99 (1), 71–92.
- Bryne, Paul (2024). “Eminent Domain Takings and Economic Development: The Effect of State Restrictions on Metropolitan Area Economic Development”. last visited Jan 25, 2025. URL: <https://papers.ssrn.com/abstract=4693907>.
- Byrne, Paul F. (2017). “Have Post-*Kelo* Restrictions on Eminent Domain Influenced State Economic Development?” *Economic Development Quarterly* 31, 81.
- Carpenter, Dick M. and John K. Ross (2010). “Do Restrictions on Eminent Domain Harm Economic Development?” *Economic Development Quarterly* 24, 337.

- Chen, Daniel L. and Susan Yeh (2023). “Growth Under the Shadow of Expropriation? The Economic Impacts of Eminent Domain”. Working paper.
- Cunningham, Chris (2013). *Estimating the Holdout Problem in Land Assembly*. Tech. rep. Working Paper 2013-19. Federal Reserve Bank of Atlanta.
- Doudchenko, Nikolay and Guido W Imbens (2016). *Balancing, Regression, Difference-In-Differences and Synthetic Control Methods: A Synthesis*. Working Paper 22791. National Bureau of Economic Research. DOI: [10.3386/w22791](https://doi.org/10.3386/w22791). URL: <http://www.nber.org/papers/w22791>.
- Ellen, Ingrid Gould, Brian J. McCabe and Eric Edward Stern (2016). *Fifty Years of Historic Preservation in New York City*. white paper. New York University Furman Center, p. 13.
- Kelo v. City of New London (2005). 545 U.S. 469.
- Fischel, William A and Perry Shapiro (1989). “A Constitutional Choice Model of Compensation for Takings”. *International Review of Law and Economics* 9 (2), 115–128.
- Goldstein v. New York State Urban Development Corp. (2009). 921 N.E.2d 164 (N.Y. 2009).
- Hoyman, Michele M. and Jamie R. McCall (2010). “Not Imminent in My Domain! County Leaders’ Attitudes toward Eminent Domain Decisions”. *Public Administration Review* 70, 885–886.
- Isaac, R. Mark, Carl Kitchens and Javier E. Portillo (2016). “Can buyer “mobility” reduce aggregation failures in land-assembly?” *Journal of Urban Economics* 95, 16–30. ISSN: 0094-1190. DOI: <https://doi.org/10.1016/j.jue.2016.06.001>. URL: <https://www.sciencedirect.com/science/article/pii/S0094119016300183>.
- Kaur v. New York State Urban Development Corp. (2010). 933 N.E.2d 721 (N.Y. 2010).
- Kerekes, Carrie B. (2011). “Government Takings: Determinants of Eminent Domain”. *Am Law Econ Rev* 13, 201.
- Kerekes, Carrie B. and Dean Stansel (2016). “Takings and Tax Revenue: Fiscal Impacts of Eminent Domain”. *Review of Law & Economics* 12, 275.
- Kitchens, Carl (2014). “The Use of Eminent Domain in Land Assembly: The Case of the Tennessee Valley Authority”. *Public Choice* 160, 455–456.
- Lanza, Steven P. et al. (2013). “The Use of Eminent Domain for Economic Development in the Era of Kelo”. *Economic Development Quarterly* 27, 352.
- Levine-Schnur, Ronit and Gary A. Wagner (2025). “Evaluation of Kelo’s Political and Economic Impact: Theory and Evidence”. *Yale Journal on Regulation* 43 (1), 219–238. URL: <https://www.yalejreg.com/print/evaluation-of-kelos-political-and-economic-impact-theory-and-evidence/>.
- Levine-Schnur, Ronit (2023). “Is the Government Exhausting Its Powers? An Empirical Examination of Eminent Domain Exercises in New York City Pre- and Post-Kelo”. *Regulation & Governance* 17, 449.
- Lopez, Edward J., R. Todd Jewell and Noel D. Campbell (2009). “Pass a Law, Any Law, Fast! State Legislative Responses to the Kelo Backlash”. *Review of Law & Economics* 5.
- Lopez, Edward J., H. Justin Pace and Jon Murphy (2025). “The Long-Term Impact of Kelo v. City of New London: Comparing State Legislative and Judicial Responses”. *Review of Law and Economics*. DOI: [10.1515/rle-2025-0005](https://doi.org/10.1515/rle-2025-0005).

- Merrill, Thomas W (1986). “The Economics of Public Use”. *Cornell Law Review* 72 (1), 61–116.
- Miceli, Thomas J and Kathleen Segerson (1994). “Regulatory Takings: When Should Compensation be Paid?” *Journal of Legal Studies* 23 (2), 749–776.
- Miceli, Thomas J and Kathleen Segerson (2007). “The Economics of Eminent Domain: Private Property, Public Use, and Just Compensation”. *Foundations and Trends in Microeconomics* 3 (4), 275–329.
- Miceli, Thomas J. and Kathleen Segerson (2021). “Assessing *Kelo*’s Legacy: Do Increased Taxes and New Jobs Justify Use of Eminent Domain?” *Journal of Real Estate Finance and Economics* 63, 161.
- Mihaly, Marc and Turner Smith (2011). “*Kelo*’s Trail: A Survey of State and Federal Legislative and Judicial Activity Five Years Later”. *Ecology L.Q.* 38, 703.
- Miller, Stephen, Gary A. Wagner and Alicia Plemmons (2025). “Employment effects of minimum wage indexing: Establishment evidence from Oregon restaurants”. *Economic Inquiry* 63 (3), 681–714. DOI: <https://doi.org/10.1111/ecin.13284>.
- Mills, Edwin S. (1967). “An Aggregative Model of Resource Allocation in a Metropolitan Area”. *American Economic Review* 57 (2), 197–210.
- Muth, Richard F. (1969). *Cities and Housing*. Chicago: University of Chicago Press.
- New York City (2007). *Glossary of Terms for Property Sales Files*. URL: https://www1.nyc.gov/assets/finance/downloads/pdf/07pdf/glossary_rsf071607.pdf.
- Otsu, Taisuke and Yoshiyasu Rai (2017). “Bootstrap inference of matching estimators for average treatment effects”. *Journal of the American Statistical Association* 112 (520), 1720–1732.
- Peters, Alan and Peter Fisher (2004). “The Failures of Economic Development Incentives”. *Journal of the American Planning Association* 70 (1), 27–37. DOI: [10.1080/01944360408976336](https://doi.org/10.1080/01944360408976336).
- Portillo, Javier E. (2019). “Land-assembly and externalities: How do positive post-development externalities affect land aggregation outcomes?” *Regional Science and Urban Economics* 77, 104–124. ISSN: 0166-0462. DOI: <https://doi.org/10.1016/j.regsciurbeco.2019.03.004>. URL: <https://www.sciencedirect.com/science/article/pii/S0166046218302382>.
- Sharp, Elaine B. and David Haider-Markel (2008). “At the Invitation of the Court: Eminent Domain Reform in State Legislatures in the Wake of the *Kelo* Decision”. *Publius: The Journal of Federalism*, 556.
- Somin, Ilya (2010). “Let There Be Blight: Blight Condemnations in New York after Goldstein and Kaur”. *Fordham Urb. LJ* 38, 1193.
- (2015). *The Grasping Hand: Kelo v. City of New London and the Limits of Eminent Domain*, p. 179.
- Turnbull, Geoffrey K. and Robert F. Salvino (2009). “Do Broader Eminent Domain Powers Increase Government Size”. *Rev. L & Econ.* 5, 785.
- Turnbull, Geoffrey K., Robert F. Salvino and Michael Tasto (2017). “Using Eminent Domain for Economic Development: Does It Increase Private Sector Employment?” *Review of Law & Economics* 14.

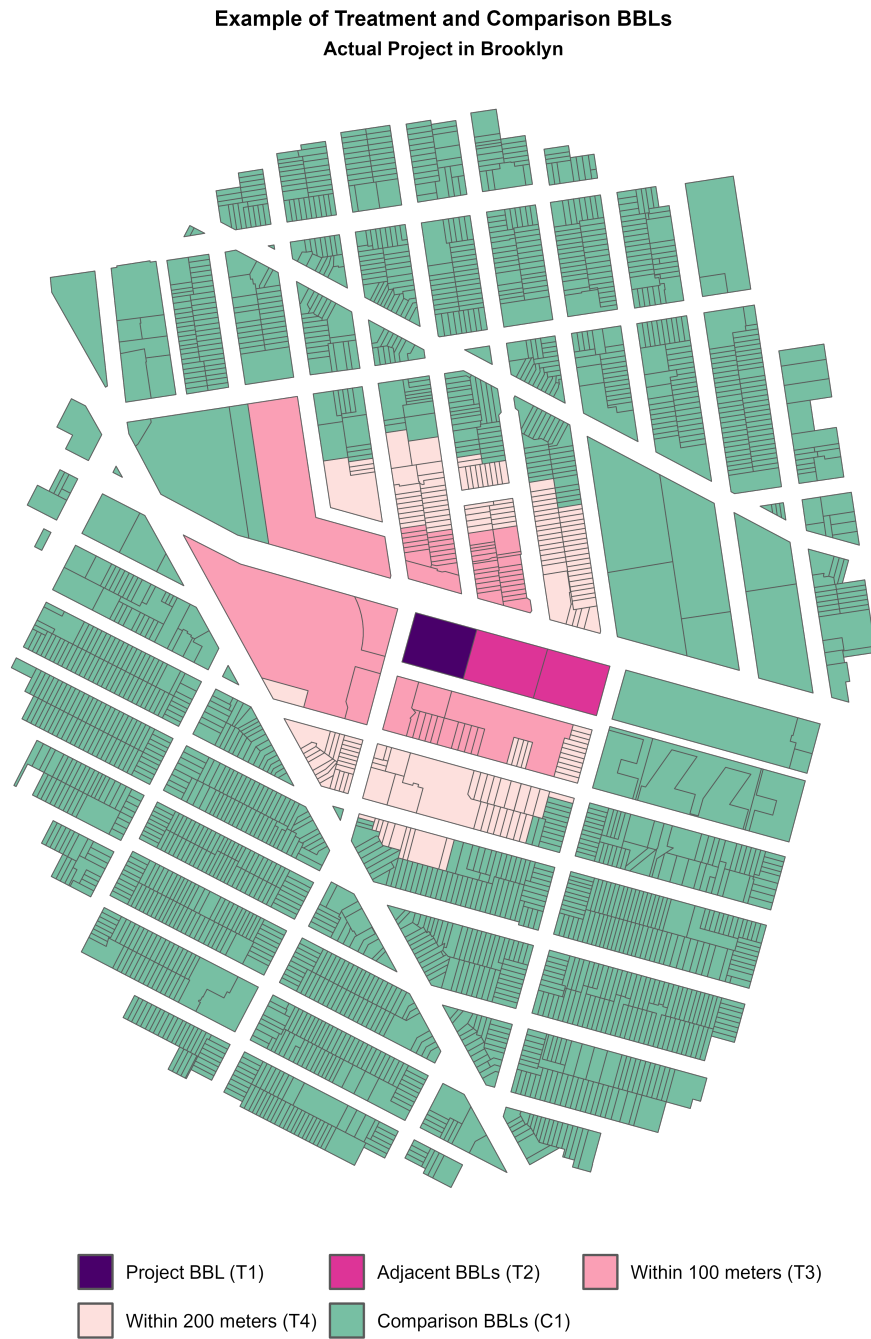
U.S. Commission on Civil Rights (2014). *The Civil Rights Implications of Eminent Domain Abuse: A Briefing Report*. Tech. rep. Washington, DC: U.S. Commission on Civil Rights.

Figure 1: Project and Borough-Block-Lot Counts by Project Type



Note: Raw data from Levine-Schnur (2023).

Figure 2: Illustration of Treated and Comparison Borough-Block-Lots



Note: This figure illustrates how the treated and comparison Borough-Block-Lots (BBLs) are selected under different scenarios. Treatment group T1 includes the project BBL. Treatment group T2 includes the project BBL *plus* the first- and second-adjacent BBLs. Treatment groups T3 and T4 expand T2 to also include any BBL that is located within 100 and 200 meters, respectively, of a project BBL. The comparison group (C1), shown in green in the figure, includes BBLs between 200 and 400 meters of the project BBL. Because the Your Economy Time Series database begins in 1998, any BBL that is located within 200 meters of any project (urban renewal or otherwise) that began before 1998 is excluded from any comparison group.

Table 1: Description of Treatment and Comparison Groups

Group Label	Description
Treatment Groups	
T1	Treatment group that includes only Borough-Block-Lots (BBLs) directly involved in urban renewal projects.
T2	Treatment group that includes renewal project BBLs from group T1 <i>plus</i> all first- and second-order adjacent BBLs.
T3	Treatment group that includes all renewal project BBLs from group T1 <i>plus</i> any BBL that is within 100 meters of a project BBL.
T4	Treatment group that includes all renewal project BBLs from group T1 <i>plus</i> any BBL that is within 200 meters of a project BBL.
Comparison Groups	
C1	BBLs located between 200 and 400 meters from a BBL directly involved in an urban renewal project.
C2	BBLs located between 200 and 600 meters from a BBL directly involved in an urban renewal project.
C3	BBLs located between 400 and 600 meters from a BBL directly involved in an urban renewal project. BBLs located between 200 and 400 meters are omitted as a buffer.
C4	BBLs located between 400 and 800 meters from a BBL directly involved in an urban renewal project. BBLs located between 200 and 400 meters are omitted as a buffer.

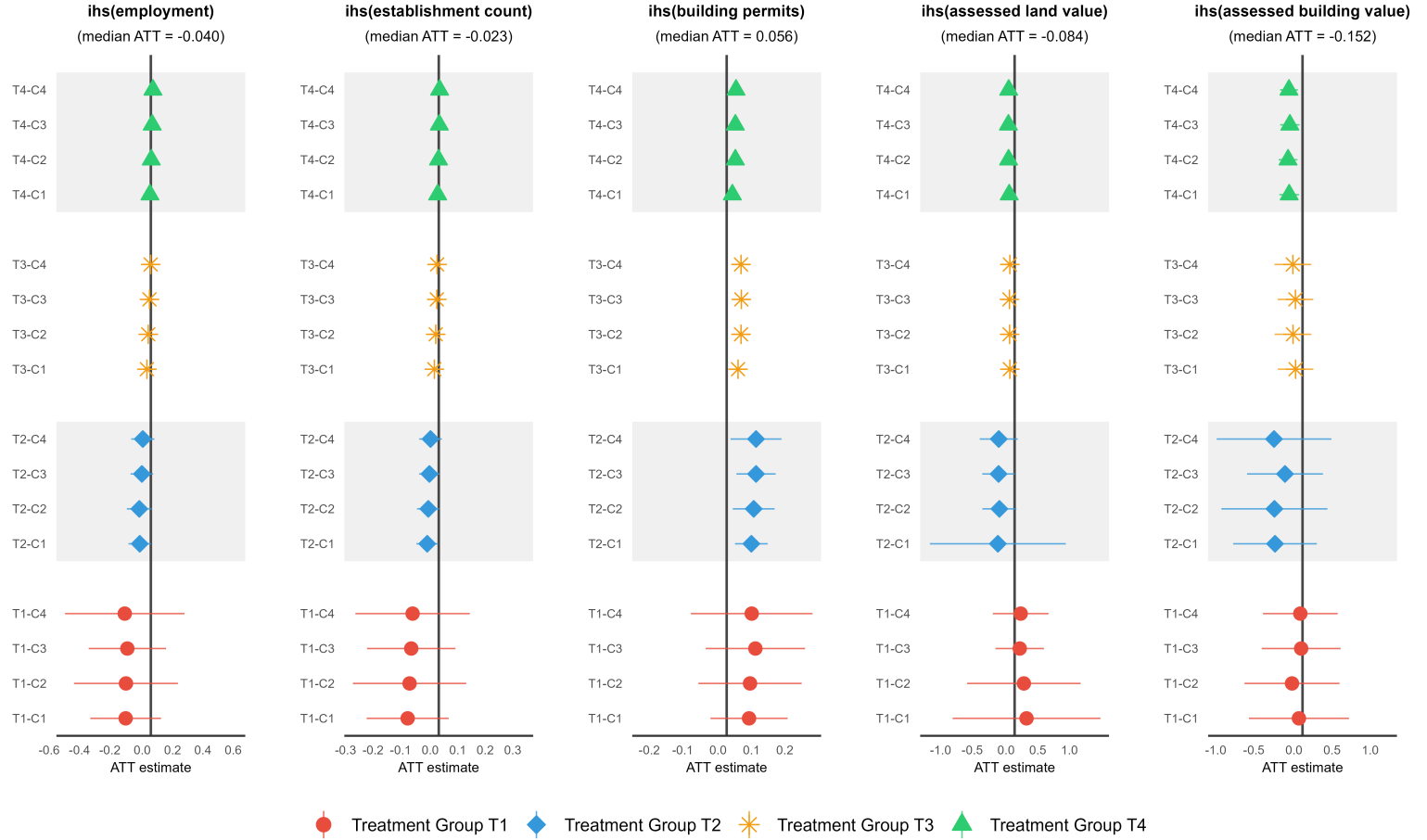
Notes: (1) Employment and business establishment data from the Your Economy Time Series database begins in 1998. As a result, urban renewal projects that were initiated prior to 1999 are omitted from the empirical analysis because we cannot observe employment or business establishment before those projects began. (2) Comparison BBLs are omitted from any comparison group if they: (a) are located within 200 meters of any non-urban renewal project or (b) if they are located within 200 meters of an urban renewal project started before 1999. (3) Figure 2 provides an illustration of the four treatment groups and the C1 comparison group using an actual urban renewal project BBL from Brooklyn.

Table 2: Descriptive Statistics and Data Sources

Variable	Group 1 (T1-C1)		Group 2 (T2-C2)		Group 3 (T3-C3)		Group 4 (T4-C4)	
	Treatment	Comparison	Treatment	Comparison	Treatment	Comparison	Treatment	Comparison
<i>Panel A: Urban Renewal Projects</i>								
employment	14.240 (91.864)	5.948 (73.596)	6.531 (55.150)	5.083 (72.778)	5.456 (49.268)	4.545 (73.305)	5.063 (39.277)	4.088 (56.039)
establishments	0.982 (1.738)	0.488 (1.569)	0.533 (1.499)	0.422 (1.483)	0.495 (1.972)	0.372 (1.416)	0.501 (1.889)	0.409 (2.618)
building permits	0.628 (3.781)	0.289 (2.339)	0.674 (7.509)	0.246 (1.398)	0.346 (2.339)	0.228 (1.363)	0.337 (2.168)	0.228 (1.399)
assessed land value	1.923 (30.544)	0.241 (14.105)	0.973 (18.027)	0.165 (8.909)	0.369 (10.503)	0.115 (0.920)	0.252 (6.483)	0.110 (1.033)
assessed building value	4.914 (72.852)	0.714 (9.160)	2.385 (42.464)	0.575 (6.449)	1.367 (26.108)	0.481 (3.739)	0.973 (16.483)	0.445 (3.393)
BBL Count (emp, estabs, permits)	122	5,403	367	13,595	1,041	7,950	2,787	18,064
Total Observations (emp, estabs, permits)	3,416	151,284	10,276	380,660	29,148	222,600	78,036	505,792
BBL Count (assess. land, build, 2002–2024)	88	5,403	160	13,595	488	7,950	1,679	18,064
Total Observations (assess. land, build, 2002–2024)	2,024	124,269	3,680	312,685	11,224	182,850	38,617	415,472
<i>Panel B: Non-Urban Renewal Projects</i>								
employment	2.994 (27.662)	9.215 (172.607)	6.970 (85.338)	6.924 (123.517)	6.133 (78.004)	5.282 (67.354)	7.398 (108.188)	4.857 (71.056)
establishments	0.311 (2.162)	0.708 (6.944)	0.550 (5.385)	0.593 (5.269)	0.538 (5.135)	0.509 (3.565)	0.598 (5.391)	0.472 (3.601)
building permits	0.184 (1.631)	0.231 (2.725)	0.211 (1.881)	0.215 (2.123)	0.204 (1.747)	0.203 (1.537)	0.223 (1.985)	0.191 (1.639)
assessed land value	0.363 (7.300)	0.228 (5.615)	0.637 (19.432)	0.197 (3.851)	0.239 (3.704)	0.176 (1.648)	0.229 (2.817)	0.170 (2.466)
assessed building value	1.133 (37.346)	0.652 (9.105)	1.473 (53.766)	0.586 (7.196)	0.738 (18.619)	0.541 (5.429)	0.739 (13.824)	0.507 (6.457)
BBL Count (emp, estabs, permits)	1,906	44,576	4,590	104,836	8,854	60,210	23,045	129,510
Total Observations (emp, estabs, permits)	53,368	1,248,128	128,520	2,935,408	247,912	1,685,880	645,260	3,626,280
BBL Count (assess. land, build, 2002–2024)	1,609	44,576	3,705	104,836	7,497	60,210	19,332	129,510
Total Observations (assess. land, build, 2002–2024)	37,007	1,025,248	85,215	2,411,228	172,431	1,384,830	444,636	2,978,730

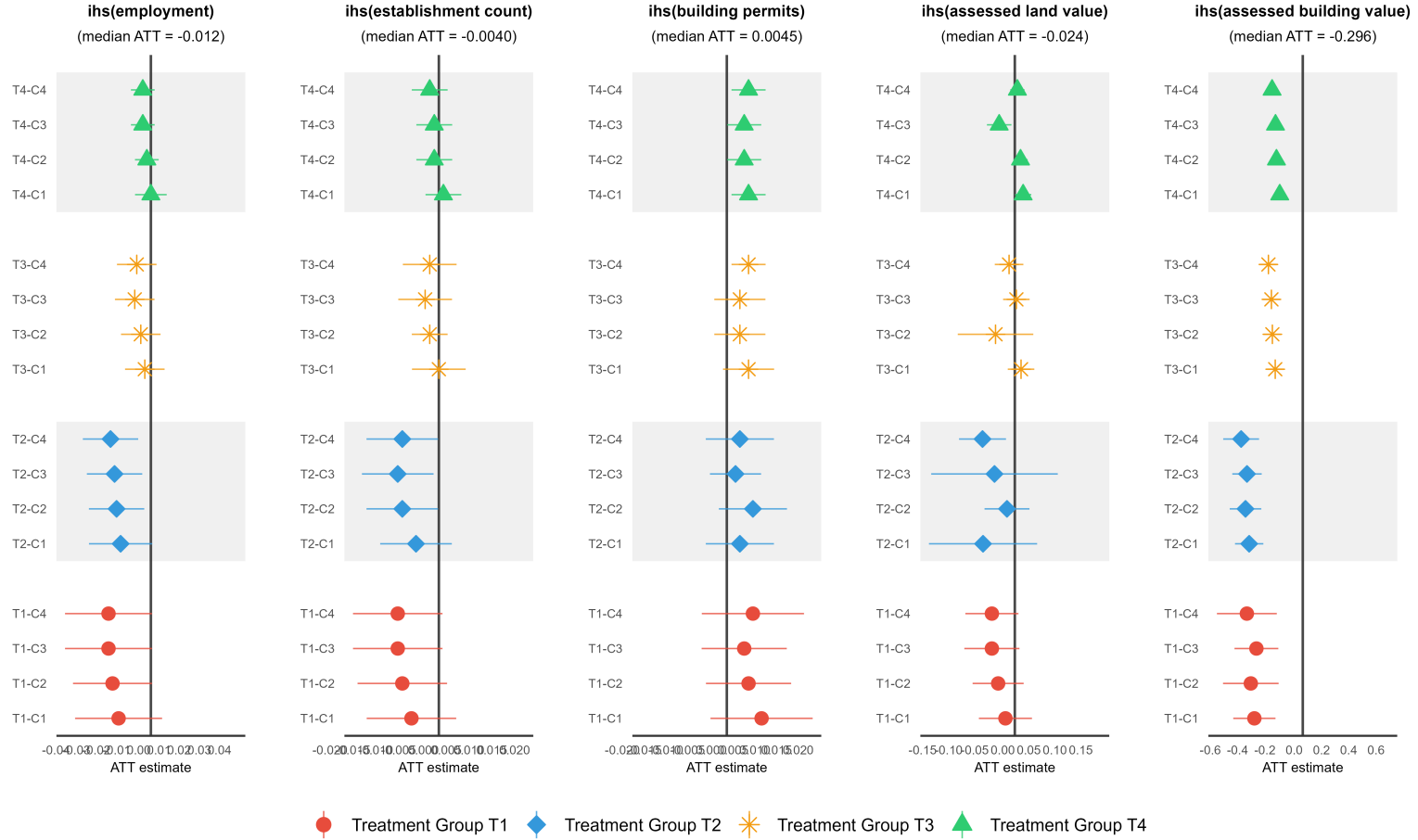
Notes: This table reports the mean and standard deviation (in parentheses) for each outcome and project type, computed over the full sample period for each BBL in the estimation sample after applying all distance-based selection criteria. The estimation sample is the same sample used in the PPSC estimation. Treatment and comparison groups are defined by distance rings. The treatment distance (T) denotes the radius (in meters) within which parcels are classified as treated; the comparison distance (C) denotes the ring from which comparison parcels are drawn. All groups exclude parcels within 200m of any other project. Employment and establishment data cover 1997–2024 and are from the Your Economy Time Series database. Building permits cover 1997–2024 and are from New York City’s Department of Buildings. Assessed land and building values cover 2002–2024, are in millions of dollars, and are from New York City MapPLUTO. BBL Count reports the number of unique tax lots; Total Observations reflects the full panel across all years for each variable group.

Figure 3: Effect of Urban Renewal Projects on Local Outcomes



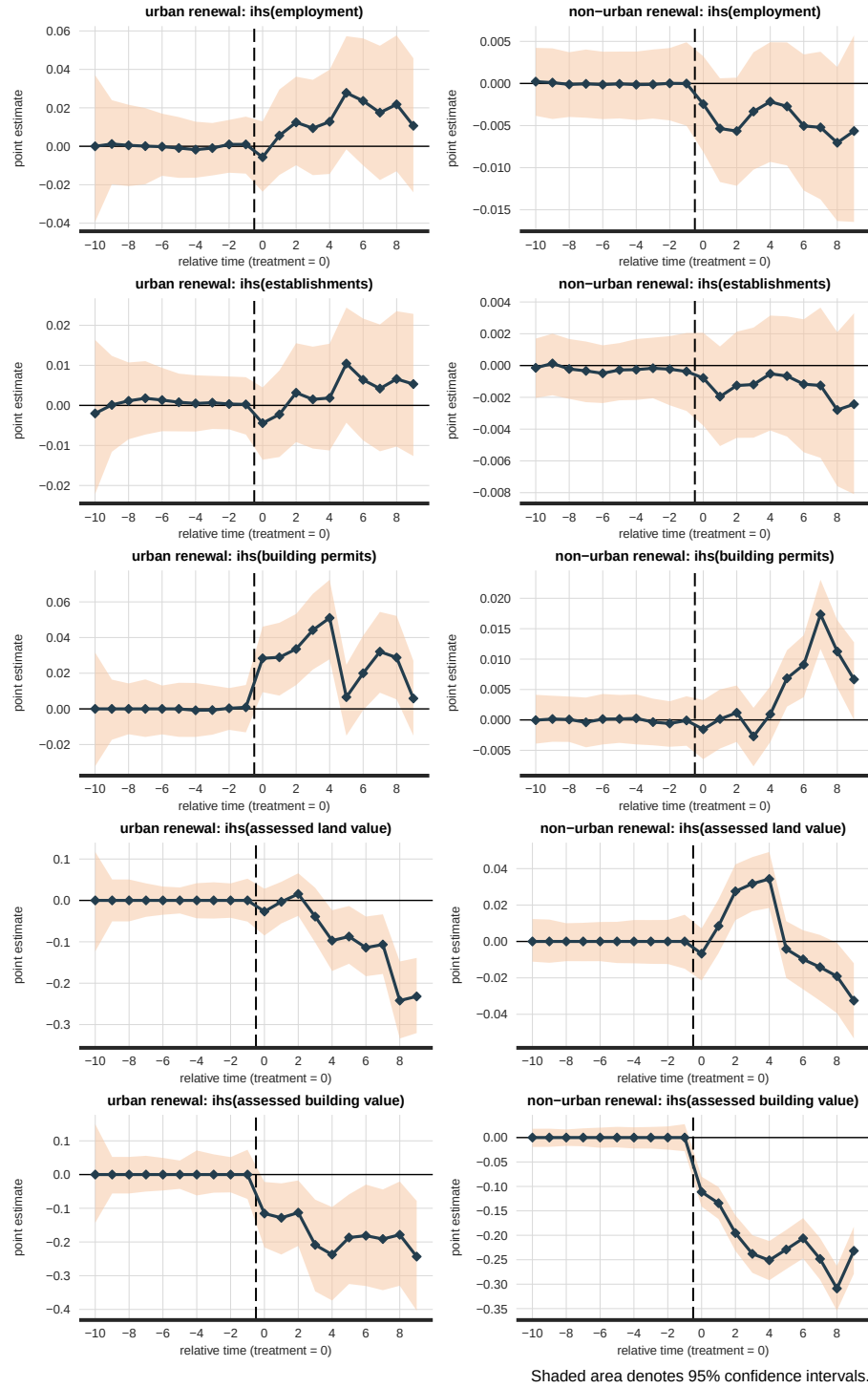
Note: Each line represents a single regression showing the ATT (identified by the marker) and 95% confidence interval. For each outcome, there are a total of 16 specifications: four treatment groups x four comparison groups. Treatment and comparison groups are defined in Table 1. *ihse* denotes the inverse hyperbolic sine transformation used to reduce dispersion. Because the partially pooled synthetic control variance estimator relies on placebo inference, confidence interval width scales with the ratio of pre-treatment residual variance to the effective number of pre-treatment periods used in the permutation distribution (Ben-Michael et al. 2022). When the number of treated units increases, the distribution of placebo estimates becomes more tightly concentrated around zero, producing more narrow confidence intervals.

Figure 4: Effect of Non-Urban Renewal Projects on Local Outcomes



Note: Each line represents a single regression showing the ATT (identified by the marker) and 95% confidence interval. For each outcome, there are a total of 16 specifications: four treatment groups x four comparison groups. Treatment and comparison groups are defined in Table 1. *ihse* denotes the inverse hyperbolic sine transformation used to reduce dispersion. Because the partially pooled synthetic control variance estimator relies on placebo inference, confidence interval width scales with the ratio of pre-treatment residual variance to the effective number of pre-treatment periods used in the permutation distribution (Ben-Michael et al. 2022). When the number of treated units increases, the distribution of placebo estimates becomes more tightly concentrated around zero, producing more narrow confidence intervals.

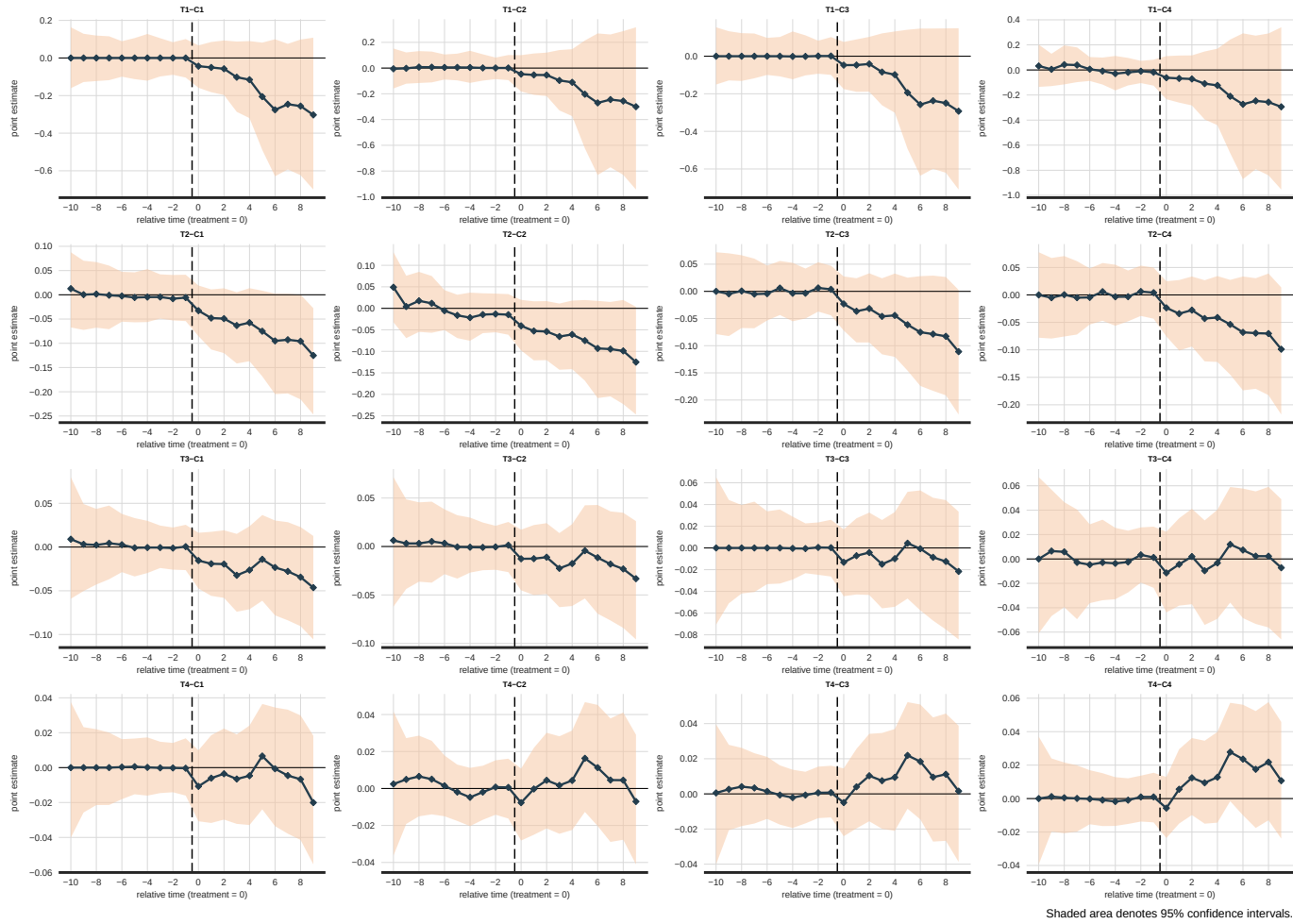
Figure 5: Selected Event Studies: Treatment Group 4 and Comparison Group 4



Note: This figure shows an event study plot for each economic outcome for urban renewal projects and non-urban renewal projects for treatment group 4 and comparison group 4. Treatment and comparison groups are defined in Table 1. ihs denotes the inverse hyperbolic sine transformation used to reduce dispersion.

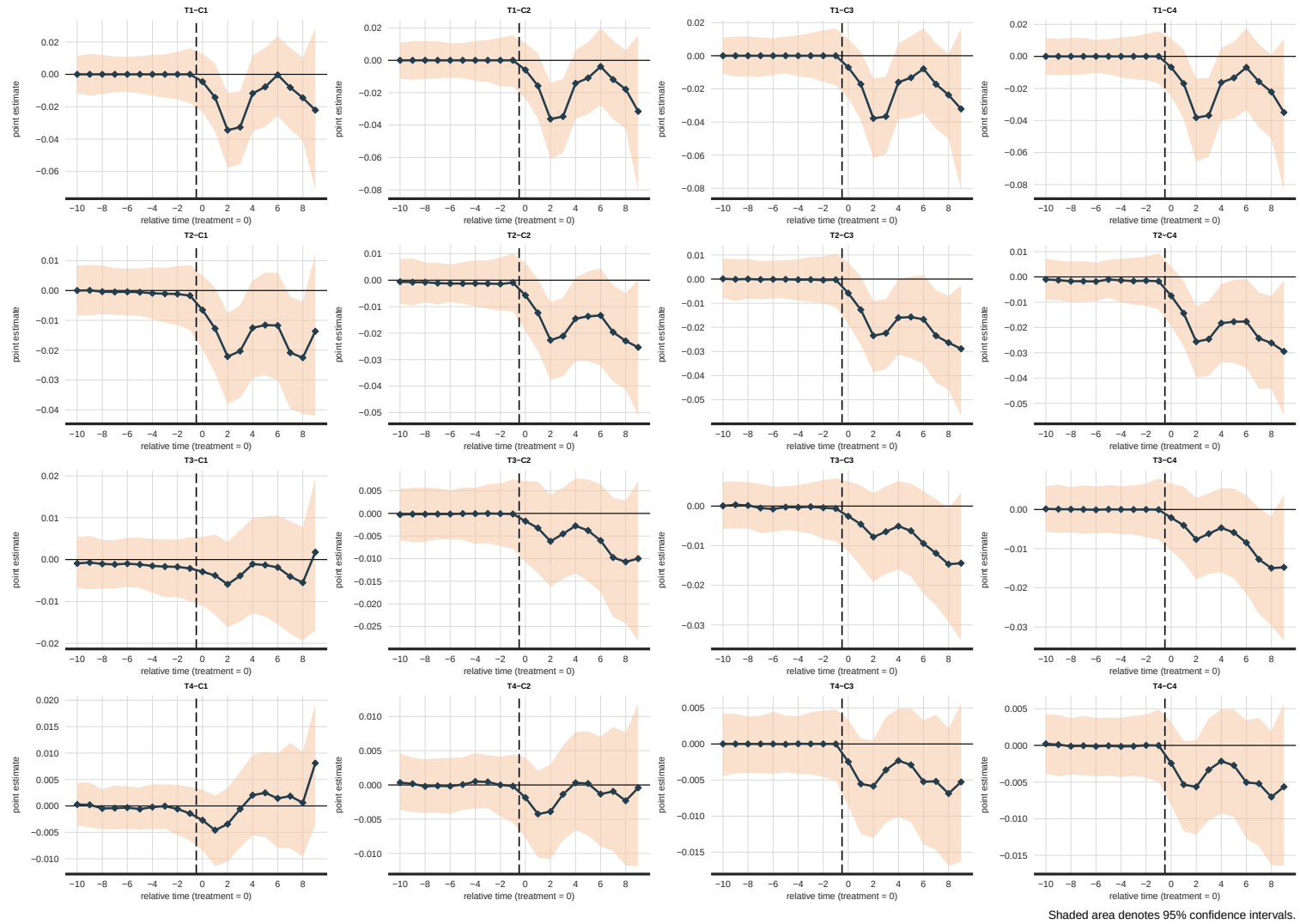
A Appendix

Figure A1: Event Studies: Urban Renewal, $\text{ih}(\text{Employment})$



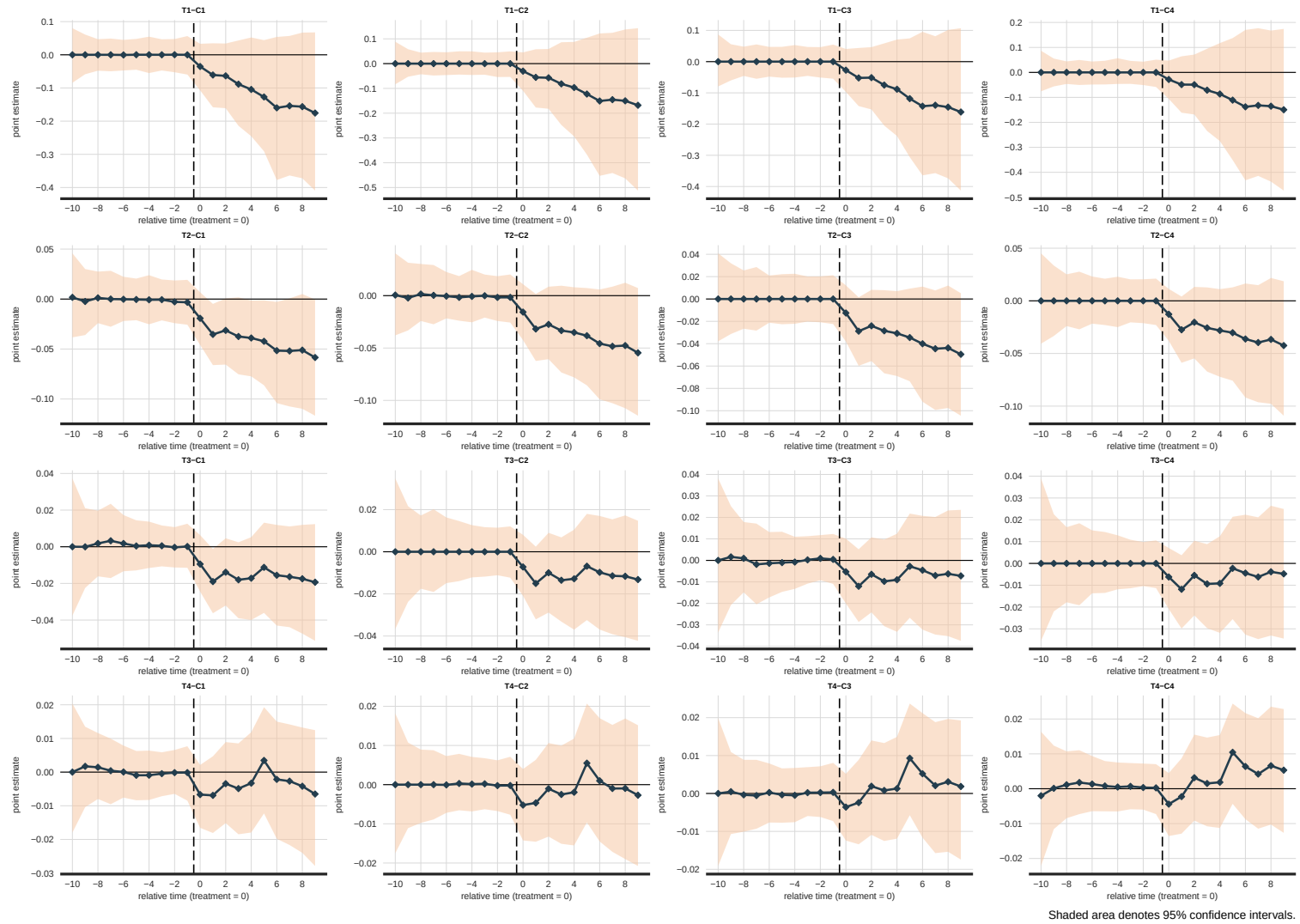
Note: This figure shows event study plots for $\text{ih}(\text{employment})$ for urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. ih denotes the inverse hyperbolic sine transformation used to reduce dispersion.

Figure A2: Event Studies: Non-Urban Renewal, $\text{ih}(\text{Employment})$



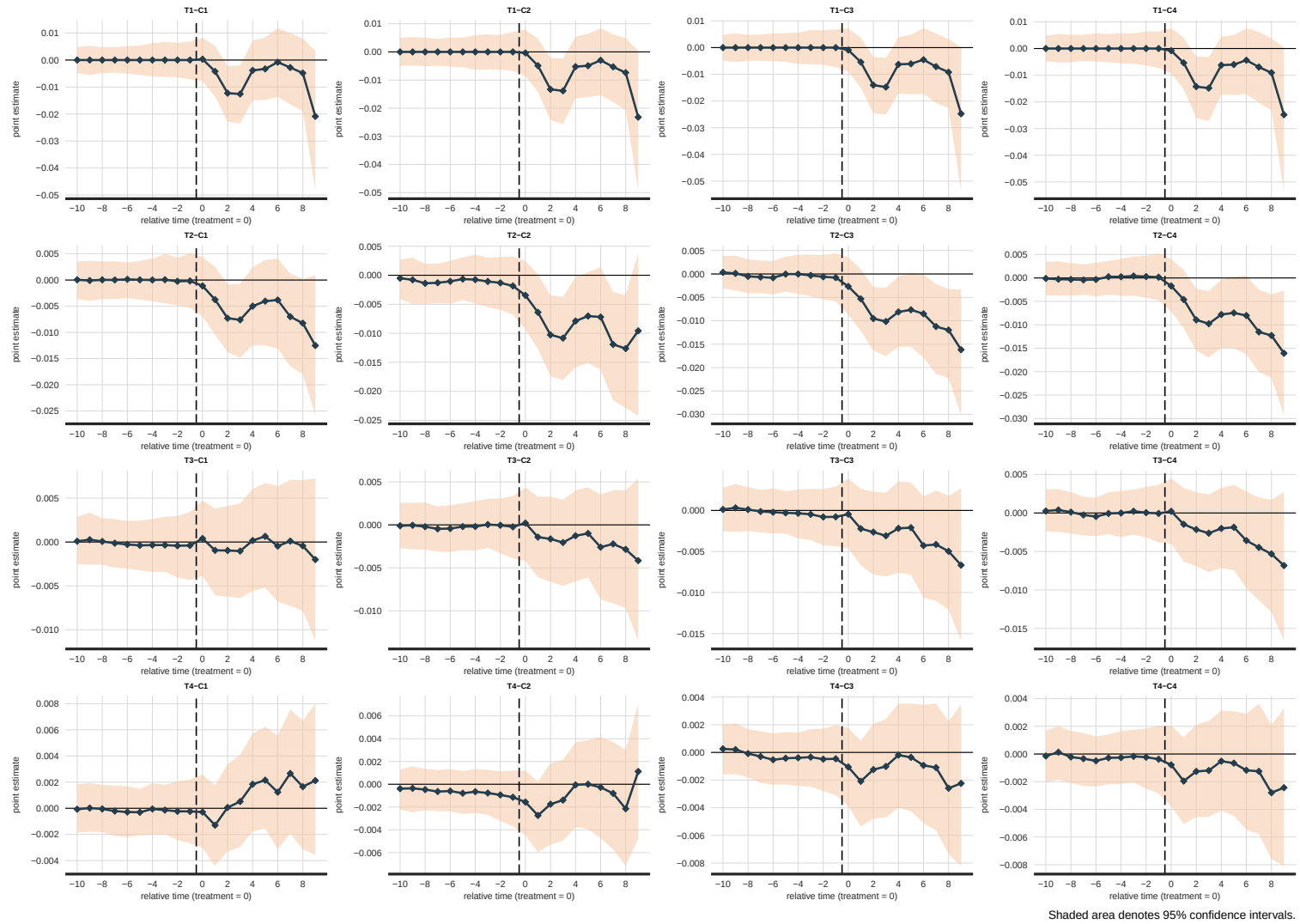
Note: This figure shows event study plots for $\text{ih}(\text{employment})$ for non-urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. ih denotes the inverse hyperbolic sine transformation used to reduce dispersion.

Figure A3: Event Studies: Urban Renewal, $\text{lhs}(\text{Establishments})$



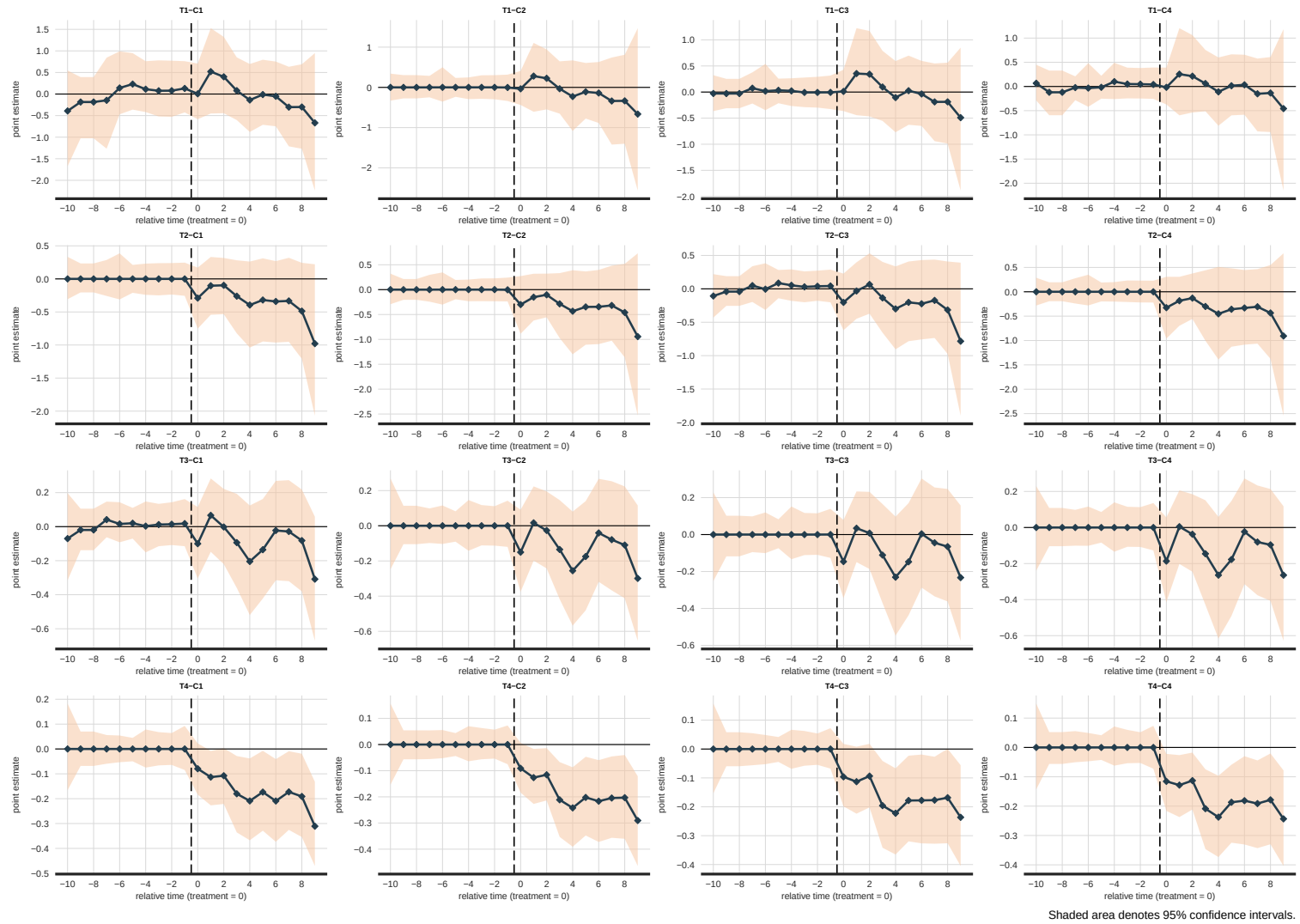
Note: This figure shows event study plots for $\text{lhs}(\text{establishments})$ for urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. lhs denotes the inverse hyperbolic sine transformation used to reduce dispersion.

Figure A4: Event Studies: Non-Urban Renewal, $\text{lhs}(\text{Establishments})$



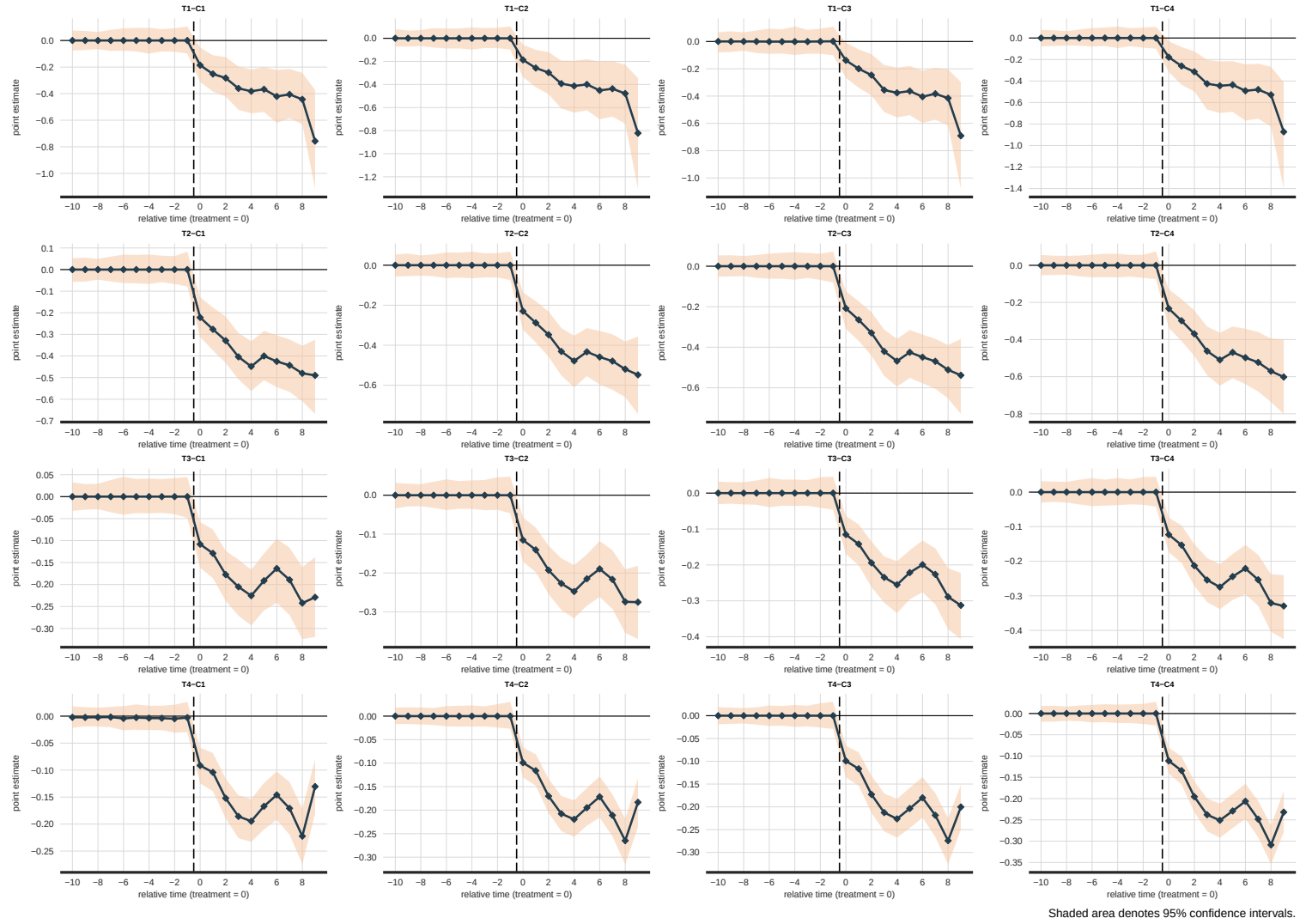
Note: This figure shows event study plots for $\text{lhs}(\text{establishments})$ for non-urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. lhs denotes the inverse hyperbolic sine transformation used to reduce dispersion.

Figure A5: Event Studies: Urban Renewal, $\text{lhs}(\text{Building Permits})$



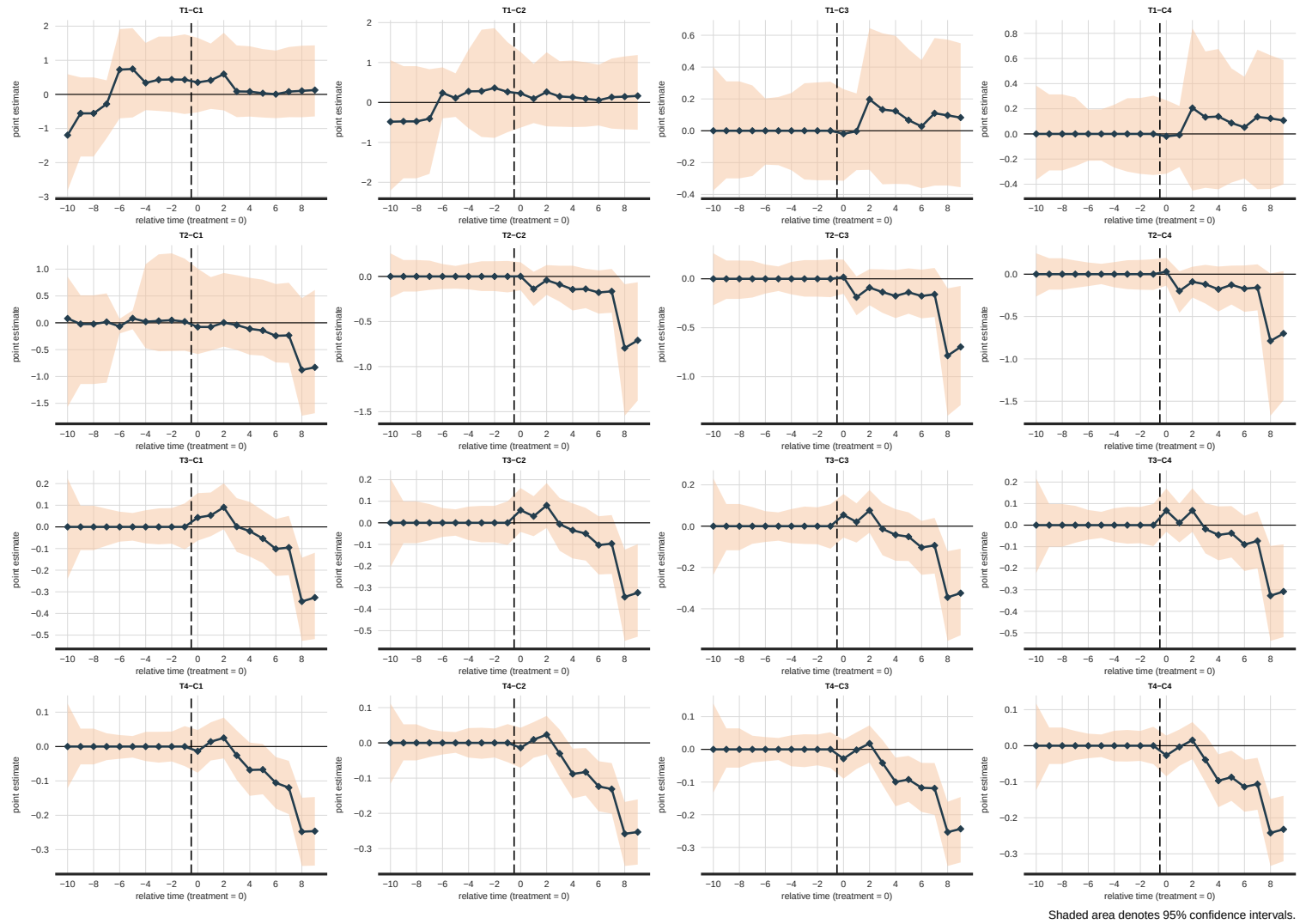
Note: This figure shows event study plots for $\text{lhs}(\text{building permits})$ for urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. lhs denotes the inverse hyperbolic sine transformation used to reduce dispersion.

Figure A6: Event Studies: Non-Urban Renewal, $\text{lhs}(\text{Building Permits})$



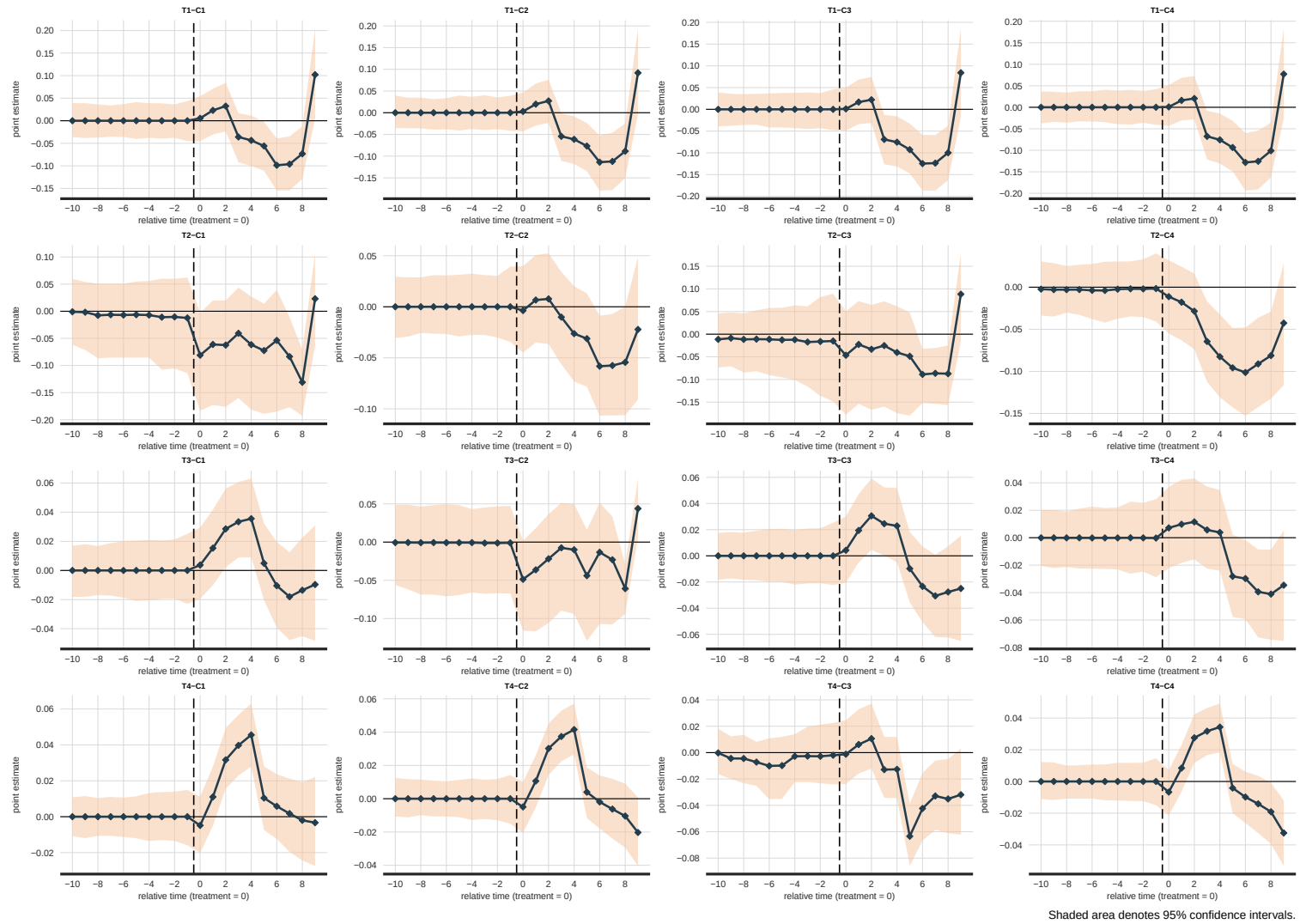
Note: This figure shows event study plots for $\text{lhs}(\text{building permits})$ for non-urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. lhs denotes the inverse hyperbolic sine transformation used to reduce dispersion.

Figure A7: Event Studies: Urban Renewal, $\ln(\text{Assessed Land Value})$



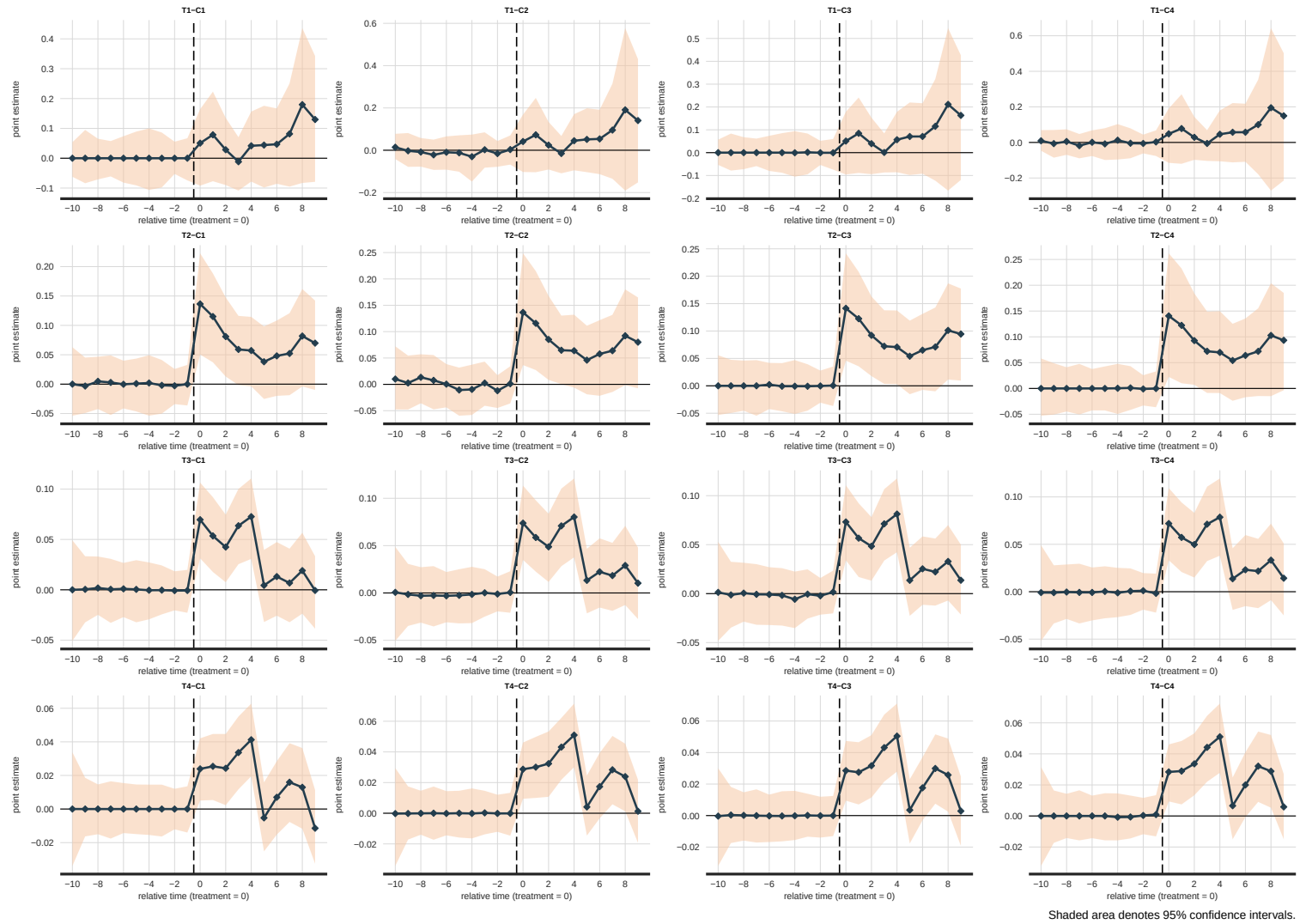
Note: This figure shows event study plots for $\ln(\text{assessed land value})$ for urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. $\ln$ denotes the inverse hyperbolic sine transformation used to reduce dispersion.

Figure A8: Event Studies: Non-Urban Renewal, $\text{lhs}(\text{Assessed Land Value})$



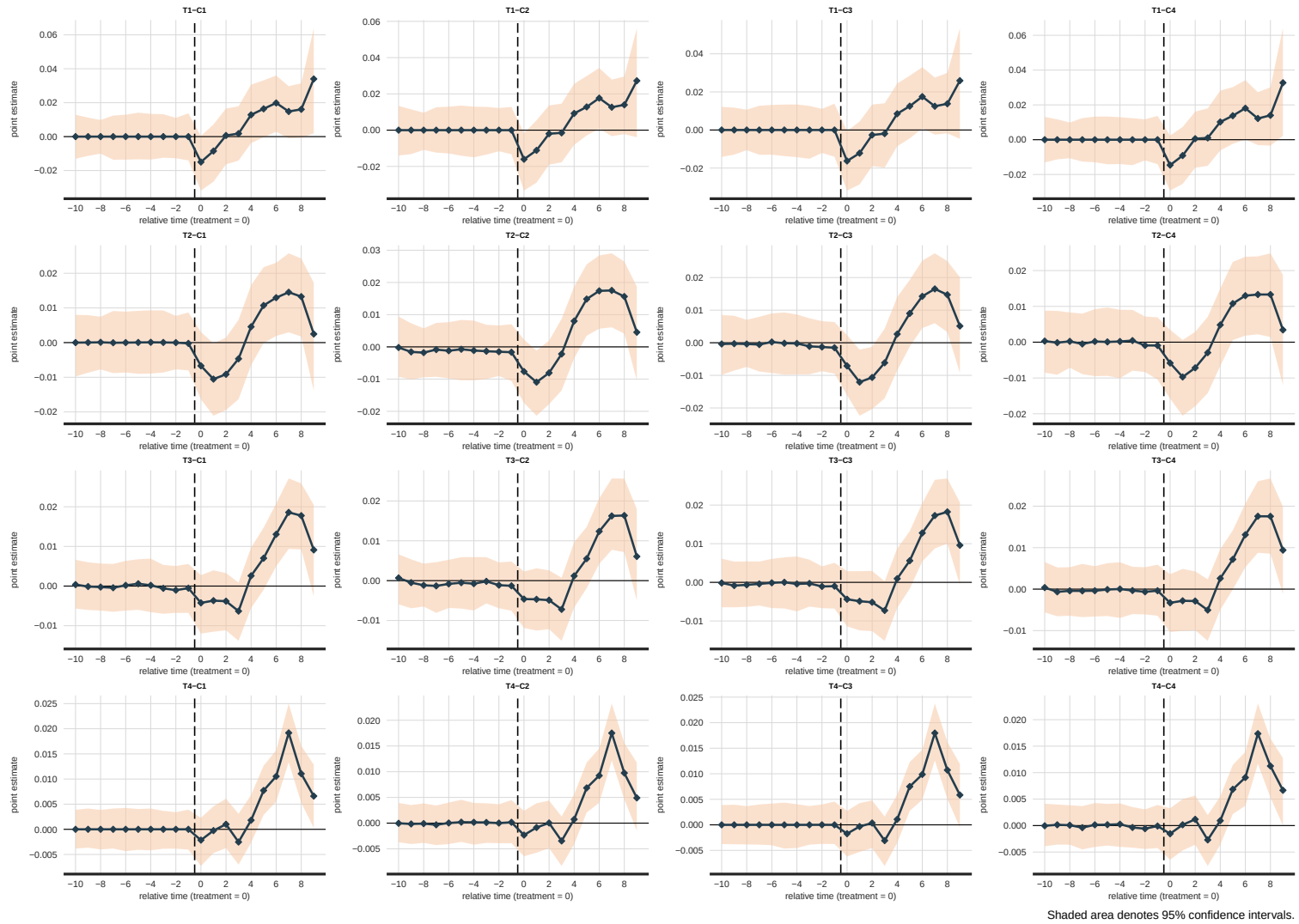
Note: This figure shows event study plots for $\text{lhs}(\text{assessed land value})$ for non-urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. lhs denotes the inverse hyperbolic sine transformation used to reduce dispersion.

Figure A9: Event Studies: Urban Renewal, $\text{lhs}(\text{Assessed Building Value})$



Note: This figure shows event study plots for $\text{lhs}(\text{assessed building value})$ for urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. lhs denotes the inverse hyperbolic sine transformation used to reduce dispersion.

Figure A10: Event Studies: Non-Urban Renewal, $\text{lhs}(\text{Assessed Building Value})$



Note: This figure shows event study plots for $\text{lhs}(\text{assessed building value})$ for non-urban renewal projects across all 16 treatment and comparison group specifications. Treatment and comparison groups are defined in Table 1. lhs denotes the inverse hyperbolic sine transformation used to reduce dispersion.